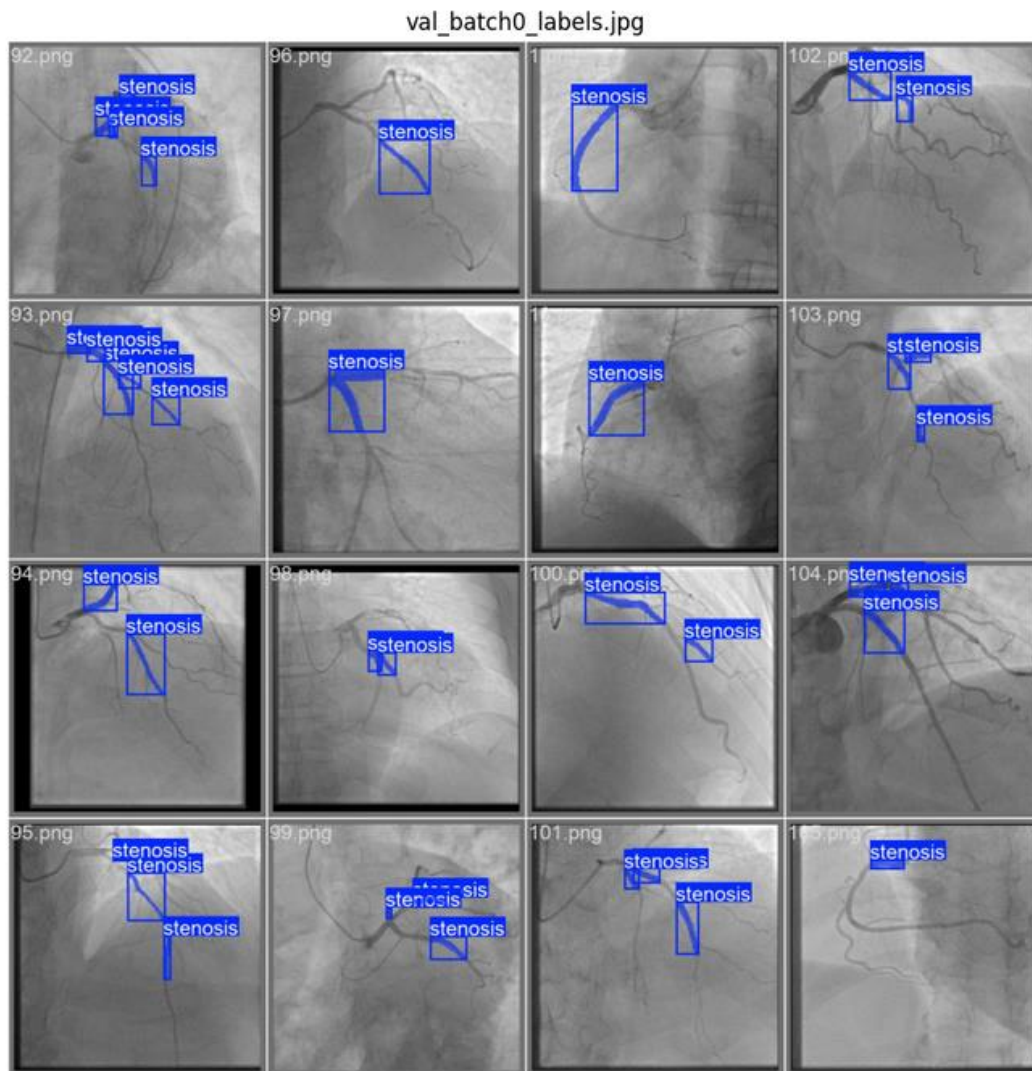


Arcade Segmentation with Yolov8



```
import numpy as np
import pandas as pd
import os

import json
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
from pathlib import Path

def load_json_annotations(json_path):
```

```

with open(json_path, 'r') as f:
    data = json.load(f)
return data

def create_mask(image_shape, segmentation):
    mask = np.zeros(image_shape, dtype=np.uint8)
    points = np.array(segmentation, dtype=np.int32).reshape(-1, 2)
    cv2.fillPoly(mask, [points], color=255)
    return mask

def visualize_segmentation(image_path, annotations, categories, image_id,
image_index):
    img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
    if img is None:
        raise ValueError(f"Could not load image at {image_path}")

    img_color = cv2.cvtColor(img, cv2.COLOR_GRAY2RGB)
    img_annotations = [ann for ann in annotations if ann['image_id'] ==
image_id]
    mask = np.zeros_like(img, dtype=np.uint8)

    for ann in img_annotations:
        category_id = ann['category_id']
        category_name = next(cat['name'] for cat in categories if cat['id']
== category_id)
        segmentation = ann['segmentation']
        ann_mask = create_mask(img.shape, segmentation)
        mask = np.maximum(mask, ann_mask)
        img_color[ann_mask == 255] = [255, 0, 0]

    plt.figure(figsize=(10, 5))
    plt.subplot(1, 2, 1)
    plt.title(f"Original Image {image_index + 1}")
    plt.imshow(img, cmap='gray')
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.title(f"Segmented Image {image_index + 1} (ID: {image_id})")
    plt.imshow(img_color)
    plt.axis('off')

    plt.show()
    return mask

def main():
    base_path = "/kaggle/input/arcade-dataset/arcade/syntax"
    train_images_path = os.path.join(base_path, "train", "images")
    train_json_path = os.path.join(base_path, "train", "annotations",
"train.json")

```

```

data = load_json_annotations(train_json_path)
images = data['images']
annotations = data['annotations']
categories = data['categories']

for idx, sample_image in enumerate(images[:5]):
    image_id = sample_image['id']
    image_file = sample_image['file_name']
    image_path = os.path.join(train_images_path, image_file)

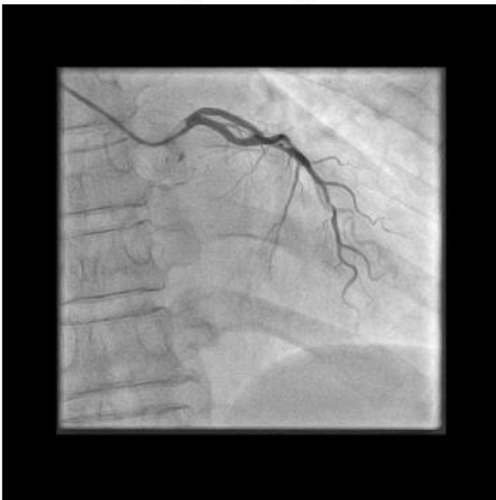
    print(f"Processing image: {image_file}")
    visualize_segmentation(image_path, annotations, categories, image_id,
idx)

if __name__ == "__main__":
    main()

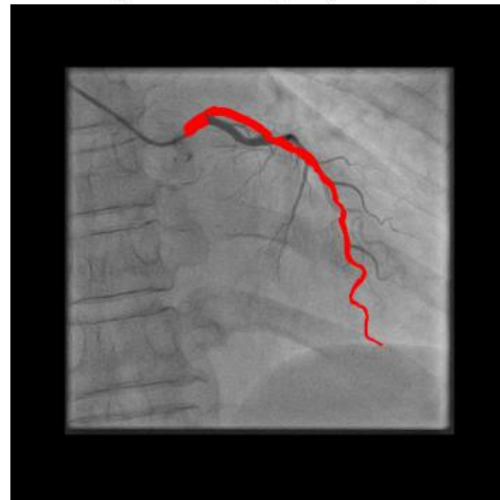
```

Processing image: 922.png

Original Image 1

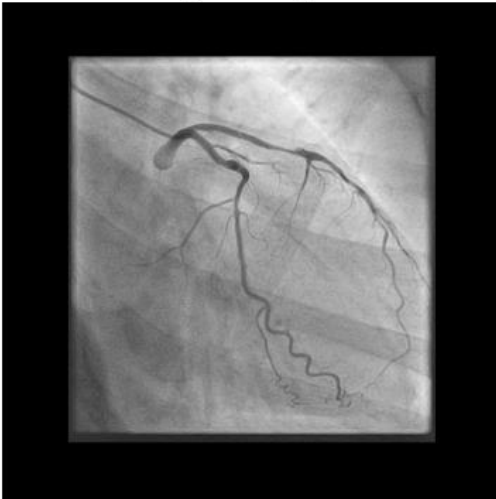


Segmented Image 1 (ID: 922)

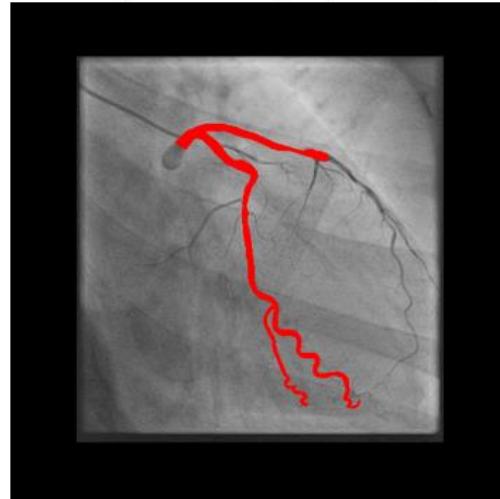


Processing image: 793.png

Original Image 2

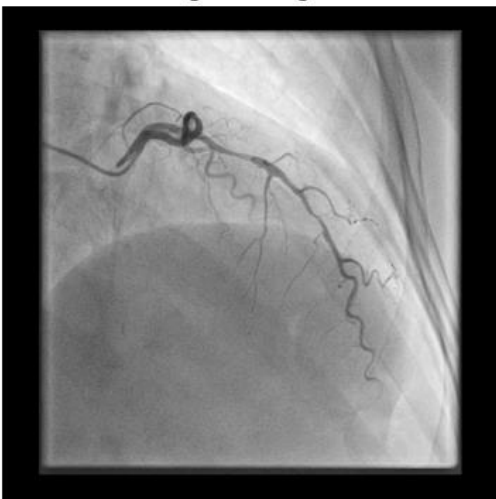


Segmented Image 2 (ID: 793)

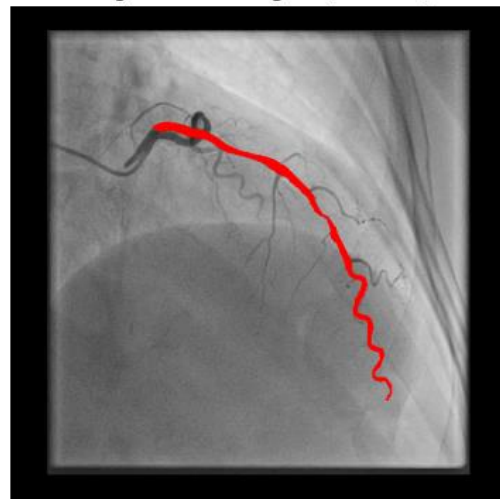


Processing image: 782.png

Original Image 3



Segmented Image 3 (ID: 782)

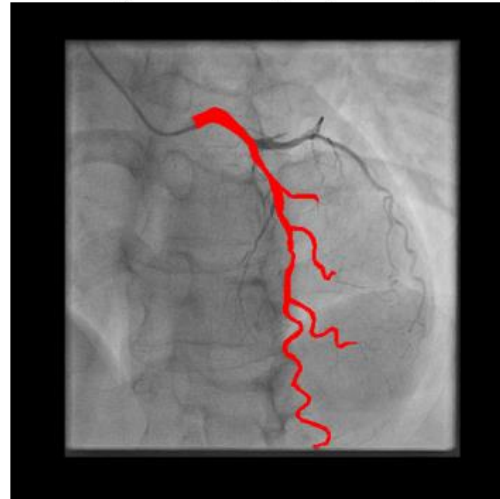


Processing image: 660.png

Original Image 4



Segmented Image 4 (ID: 660)

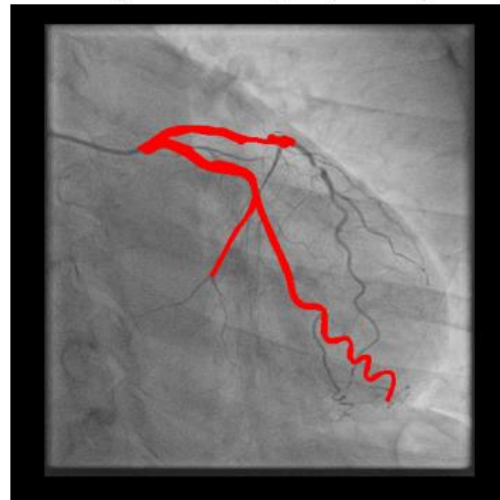


Processing image: 708.png

Original Image 5



Segmented Image 5 (ID: 708)



```
!pip install ultralytics pycocotools
```

```
import os
import shutil
import json
from ultralytics import YOLO

dataset_path = '/kaggle/input/arcade-dataset/arcade/stenosis'
work_dir = '/kaggle/working'
yolo_data_dir = os.path.join(work_dir, 'yolov8_data')
train_labels_dir = os.path.join(yolo_data_dir, 'train', 'labels')
val_labels_dir = os.path.join(yolo_data_dir, 'val', 'labels')
test_labels_dir = os.path.join(yolo_data_dir, 'test', 'labels')
```

```

os.makedirs(os.path.join(yolo_data_dir, 'train', 'images'), exist_ok=True)
os.makedirs(os.path.join(yolo_data_dir, 'train', 'labels'), exist_ok=True)
os.makedirs(os.path.join(yolo_data_dir, 'val', 'images'), exist_ok=True)
os.makedirs(os.path.join(yolo_data_dir, 'val', 'labels'), exist_ok=True)
os.makedirs(os.path.join(yolo_data_dir, 'test', 'images'), exist_ok=True)
os.makedirs(os.path.join(yolo_data_dir, 'test', 'labels'), exist_ok=True)

```

```

shutil.copytree(os.path.join(dataset_path, 'train', 'images'),
os.path.join(yolo_data_dir, 'train', 'images'), dirs_exist_ok=True)
shutil.copytree(os.path.join(dataset_path, 'val', 'images'),
os.path.join(yolo_data_dir, 'val', 'images'), dirs_exist_ok=True)
shutil.copytree(os.path.join(dataset_path, 'test', 'images'),
os.path.join(yolo_data_dir, 'test', 'images'), dirs_exist_ok=True)

```

```

print("Images copied to YOLO-formatted directory structure.")

```

```

def convert_coco_to_yolo_seg(json_file_path, output_dir):

```

```

    """

```

```

    Converts COCO-style JSON annotations to YOLOv8 segmentation format.

```

```

    Args:

```

```

        json_file_path (str): Path to the COCO JSON annotation file.

```

```

        output_dir (str): Path to the directory where YOLO annotation files
will be saved.

```

```

    """

```

```

    os.makedirs(output_dir, exist_ok=True)

```

```

    with open(json_file_path, 'r') as f:
        data = json.load(f)

```

```

    image_info = {img['id']: {'file_name': img['file_name'], 'width':
img['width'], 'height': img['height']}} for img in data['images']]

```

```

    processed_annotations = 0
    skipped_annotations = 0

```

```

    for ann in data['annotations']:
        image_id = ann['image_id']
        class_id = 0

```

```

        if 'segmentation' not in ann or not ann['segmentation'] or not
ann['segmentation'][0]:

```

```

            skipped_annotations += 1
            continue

```

```

        segmentation = ann['segmentation'][0]

```

```

        if len(segmentation) % 2 != 0 or len(segmentation) == 0:
            skipped_annotations += 1
            continue

```

```

img_data = image_info.get(image_id)
if not img_data:
    skipped_annotations += 1
    continue

img_width = img_data['width']
img_height = img_data['height']

normalized_coords = []
for i in range(0, len(segmentation), 2):
    x = segmentation[i] / img_width
    y = segmentation[i+1] / img_height
    normalized_coords.append(f"{x:.6f} {y:.6f}")

if normalized_coords:
    yolo_line = f"{class_id} " + " ".join(normalized_coords)
    file_name = os.path.splitext(img_data['file_name'])[0]
    output_file = os.path.join(output_dir, f"{file_name}.txt")

    with open(output_file, 'a') as f:
        f.write(yolo_line + "\n")
    processed_annotations += 1
else:
    skipped_annotations += 1

print(f"Conversion complete. Processed {processed_annotations}
annotations and skipped {skipped_annotations}.")
print(f"YOLO annotations saved to {output_dir}")

train_json_path = os.path.join(dataset_path, 'train', 'annotations',
'train.json')
val_json_path = os.path.join(dataset_path, 'val', 'annotations', 'val.json')
test_json_path = os.path.join(dataset_path, 'test', 'annotations',
'test.json')

print("Starting COCO to YOLO conversion for training data...")
convert_coco_to_yolo_seg(train_json_path, train_labels_dir)
print(f"Generated {len(os.listdir(train_labels_dir))} label files for
training.")

print("\nStarting COCO to YOLO conversion for validation data...")
convert_coco_to_yolo_seg(val_json_path, val_labels_dir)
print(f"Generated {len(os.listdir(val_labels_dir))} label files for
validation.")

print("\nStarting COCO to YOLO conversion for test data...")
convert_coco_to_yolo_seg(test_json_path, test_labels_dir)
print(f"Generated {len(os.listdir(test_labels_dir))} label files for

```



```
testing.")
```

```
data_yaml_content = f"""
path: {yolo_data_dir}
train: train/images
val: val/images
test: test/images
names:
  - stenosis
nc: 1
"""
```

Collecting ultralytics

```
Downloading ultralytics-8.3.193-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: pycocotools in /usr/local/lib/python3.11/dist-packages (2.0.10)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (3.7.2)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (4.11.0.86)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (11.2.1)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (6.0.2)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (1.15.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (2.6.0+cu124)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics) (0.21.0+cu124)
Requirement already satisfied: psutil in /usr/local/lib/python3.11/dist-packages (from ultralytics) (7.0.0)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.11/dist-packages (from ultralytics) (9.0.0)
Requirement already satisfied: polars in /usr/local/lib/python3.11/dist-packages (from ultralytics) (1.21.0)
Collecting ultralytics-thop>=2.0.0 (from ultralytics)
  Downloading ultralytics_thop-2.0.17-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.2)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralytics)
```


(4.58.4)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralalytics)
(1.4.8)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralalytics)
(25.0)
Requirement already satisfied: pyparsing<3.1,>=2.3.1 in
/usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralalytics)
(3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.11/dist-packages (from matplotlib>=3.3.0->ultralalytics)
(2.9.0.post0)
Requirement already satisfied: mkl_fft in /usr/local/lib/python3.11/dist-
packages (from numpy>=1.23.0->ultralalytics) (1.3.8)
Requirement already satisfied: mkl_random in /usr/local/lib/python3.11/dist-
packages (from numpy>=1.23.0->ultralalytics) (1.2.4)
Requirement already satisfied: mkl_umath in /usr/local/lib/python3.11/dist-
packages (from numpy>=1.23.0->ultralalytics) (0.1.1)
Requirement already satisfied: mkl in /usr/local/lib/python3.11/dist-packages
(from numpy>=1.23.0->ultralalytics) (2025.2.0)
Requirement already satisfied: tbb4py in /usr/local/lib/python3.11/dist-
packages (from numpy>=1.23.0->ultralalytics) (2022.2.0)
Requirement already satisfied: mkl-service in /usr/local/lib/python3.11/dist-
packages (from numpy>=1.23.0->ultralalytics) (2.4.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.23.0->ultralalytics)
(3.4.2)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.23.0->ultralalytics)
(3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.23.0->ultralalytics)
(2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.11/dist-packages (from requests>=2.23.0->ultralalytics)
(2025.6.15)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-
packages (from torch>=1.8.0->ultralalytics) (3.18.0)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralalytics)
(4.14.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-
packages (from torch>=1.8.0->ultralalytics) (3.5)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-
packages (from torch>=1.8.0->ultralalytics) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-
packages (from torch>=1.8.0->ultralalytics) (2025.5.1)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch>=1.8.0->ultralalytics)
 Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-

manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Collecting nvidia-cublas-cu12==12.4.5.8 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cufft-cu12==11.2.1.3 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-curand-cu12==10.3.5.147 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Collecting nvidia-cuspars-cu12==12.3.1.170 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_cuspars-cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralytics) (0.6.2)
 Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralytics) (2.21.5)
 Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralytics) (12.4.127)
 Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch>=1.8.0->ultralytics)
 Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralytics) (3.2.0)
 Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch>=1.8.0->ultralytics) (1.13.1)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch>=1.8.0->ultralytics) (1.3.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
 Requirement already satisfied: MarkupSafe>=2.0 in

```

/usr/local/lib/python3.11/dist-packages (from jinja2->torch>=1.8.0-
>ultralalytics) (3.0.2)
Requirement already satisfied: intel-openmp<2026,>=2024 in
/usr/local/lib/python3.11/dist-packages (from mkl->numpy>=1.23.0-
>ultralalytics) (2024.2.0)
Requirement already satisfied: tbb==2022.* in /usr/local/lib/python3.11/dist-
packages (from mkl->numpy>=1.23.0->ultralalytics) (2022.2.0)
Requirement already satisfied: tcmlib==1.* in /usr/local/lib/python3.11/dist-
packages (from tbb==2022.*->mkl->numpy>=1.23.0->ultralalytics) (1.4.0)
Requirement already satisfied: intel-cmplr-lib-rt in
/usr/local/lib/python3.11/dist-packages (from mkl_umath->numpy>=1.23.0-
>ultralalytics) (2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.11/dist-packages (from intel-openmp<2026,>=2024->mkl-
>numpy>=1.23.0->ultralalytics) (2024.2.0)
Downloading ultralytics-8.3.193-py3-none-any.whl (1.1 MB)
-----1.1/1.1 MB 19.2 MB/s eta 0:00:00a
0:00:01
anylinux2014_x86_64.whl (363.4 MB)
-----363.4/363.4 MB 4.8 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (13.8 MB)
-----13.8/13.8 MB 93.8 MB/s eta
0:00:00:00:010:01
anylinux2014_x86_64.whl (24.6 MB)
-----24.6/24.6 MB 65.9 MB/s eta
0:00:00:00:0100:01
e_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (883 kB)
-----883.7/883.7 kB 53.1 MB/s eta 0:00:00
anylinux2014_x86_64.whl (664.8 MB)
-----664.8/664.8 MB 2.5 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (211.5 MB)
-----211.5/211.5 MB 8.1 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (56.3 MB)
-----56.3/56.3 MB 30.9 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (127.9 MB)
-----127.9/127.9 MB 8.7 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (207.5 MB)
-----207.5/207.5 MB 8.2 MB/s eta
0:00:00:00:0100:01
anylinux2014_x86_64.whl (21.1 MB)
-----21.1/21.1 MB 72.4 MB/s eta
0:00:00:00:0100:01
e-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12,
nvidia-cusparse-cu12, nvidia-cudnn-cu12, nvidia-cusolver-cu12, ultralytics-
thop, ultralytics

```

Attempting uninstall: nvidia-nvjitlink-cu12
Found existing installation: nvidia-nvjitlink-cu12 12.5.82
Uninstalling nvidia-nvjitlink-cu12-12.5.82:
Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82
Attempting uninstall: nvidia-curand-cu12
Found existing installation: nvidia-curand-cu12 10.3.6.82
Uninstalling nvidia-curand-cu12-10.3.6.82:
Successfully uninstalled nvidia-curand-cu12-10.3.6.82
Attempting uninstall: nvidia-cufft-cu12
Found existing installation: nvidia-cufft-cu12 11.2.3.61
Uninstalling nvidia-cufft-cu12-11.2.3.61:
Successfully uninstalled nvidia-cufft-cu12-11.2.3.61
Attempting uninstall: nvidia-cuda-runtime-cu12
Found existing installation: nvidia-cuda-runtime-cu12 12.5.82
Uninstalling nvidia-cuda-runtime-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82
Attempting uninstall: nvidia-cuda-nvrtc-cu12
Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82
Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82
Attempting uninstall: nvidia-cuda-cupti-cu12
Found existing installation: nvidia-cuda-cupti-cu12 12.5.82
Uninstalling nvidia-cuda-cupti-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82
Attempting uninstall: nvidia-cublas-cu12
Found existing installation: nvidia-cublas-cu12 12.5.3.2
Uninstalling nvidia-cublas-cu12-12.5.3.2:
Successfully uninstalled nvidia-cublas-cu12-12.5.3.2
Attempting uninstall: nvidia-cusparse-cu12
Found existing installation: nvidia-cusparse-cu12 12.5.1.3
Uninstalling nvidia-cusparse-cu12-12.5.1.3:
Successfully uninstalled nvidia-cusparse-cu12-12.5.1.3
Attempting uninstall: nvidia-cudnn-cu12
Found existing installation: nvidia-cudnn-cu12 9.3.0.75
Uninstalling nvidia-cudnn-cu12-9.3.0.75:
Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75
Attempting uninstall: nvidia-cusolver-cu12
Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Successfully installed nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.2.1.3 nvidia-curand-cu12-10.3.5.147 nvidia-cusolver-cu12-11.6.1.9 nvidia-cusparse-cu12-12.3.1.170 nvidia-nvjitlink-cu12-12.4.127 ultralytics-8.3.193 ultralytics-thop-2.0.17
Creating new Ultralytics Settings v0.0.6 file ✓
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see

<https://docs.ultralytics.com/quickstart/#ultralytics-settings>.
Images copied to YOLO-formatted directory structure.
Starting COCO to YOLO conversion for training data...
Conversion complete. Processed 1625 annotations and skipped 0.
YOLO annotations saved to /kaggle/working/yolov8_data/train/labels
Generated 997 label files for training.

Starting COCO to YOLO conversion for validation data...
Conversion complete. Processed 406 annotations and skipped 0.
YOLO annotations saved to /kaggle/working/yolov8_data/val/labels
Generated 200 label files for validation.

Starting COCO to YOLO conversion for test data...
Conversion complete. Processed 386 annotations and skipped 0.
YOLO annotations saved to /kaggle/working/yolov8_data/test/labels
Generated 300 label files for testing.

```
with open(os.path.join(work_dir, 'data.yaml'), 'w') as f:  
    f.write(data_yaml_content)
```

```
model = YOLO('yolov8n-seg.pt')
```

```
model.train(  
    data=os.path.join(work_dir, 'data.yaml'),  
    epochs=50,  
    imgsz=640,  
    batch=16,  
    device=0,  
    project=os.path.join(work_dir, 'runs'),  
    name='stenosis_segmentation',  
    patience=100,  
    verbose=True,  
    task='segment'  
)
```

```
model.val(data=os.path.join(work_dir, 'data.yaml'))
```

```
results = model.predict(  
    source=os.path.join(dataset_path, 'test', 'images'),  
    save=True,  
    project=os.path.join(work_dir, 'runs'),  
    name='test_predictions',  
    conf=0.25,  
    iou=0.7  
)
```

```
print("Training and inference completed. Results saved in  
/kaggle/working/runs.")
```

/ultralytics/assets/releases/download/v8.3.0/yolov8n-seg.pt to 'yolov8n-seg.pt': 100% ————— 6.7MB 80.9MB/s 0.1s

Ultralytics 8.3.193 Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/kaggle/working/data.yaml, degrees=0.0, deterministic=True, device=0, df1=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=50, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8n-seg.pt, momentum=0.937, mosaic=1.0, multi_scale=False, name=stenosis_segmentation, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=/kaggle/working/runs, rect=False, resume=False, retina_masks=False, save=True, save_conf=False, save_crop=False, save_dir=/kaggle/working/runs/stenosis_segmentation, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=segment, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ————— 755.1KB 15.8MB/s 0.0s

Overriding model.yaml nc=80 with nc=1

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	7360	ultralytics.nn.modules.block.C2f
[32, 32, 1, True]				
3	-1	1	18560	ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]				
4	-1	2	49664	ultralytics.nn.modules.block.C2f
[64, 64, 2, True]				
5	-1	1	73984	ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]				
6	-1	2	197632	ultralytics.nn.modules.block.C2f
[128, 128, 2, True]				
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	460288	ultralytics.nn.modules.block.C2f

```

[256, 256, 1, True]
  9          -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10          -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11          [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12          -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
 13          -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14          [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15          -1  1    37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16          -1  1    36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17          [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18          -1  1   123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19          -1  1   147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20          [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21          -1  1   493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22          [15, 18, 21]  1  1004275  ultralytics.nn.modules.head.Segment
[1, 32, 64, [64, 128, 256]]
YOLOv8n-seg summary: 151 layers, 3,263,811 parameters, 3,263,795 gradients,
11.5 GFLOPs

```

```

Transferred 381/417 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to 'yolo11n.pt': 100%
-----5.4MB 69.3MB/s 0.1s
AMP: checks passed ✓
train: Fast image access ✓(ping: 0.0±0.0 ms, read: 2451.8±691.4 MB/s, size:
125.9 KB)
train: Scanning /kaggle/working/yolov8_data/train/labels... 997 images, 3
backgrounds, 0 corrupt: 100% -----1000/1000 1.1Kit/s 0.9s.1ss
train: New cache created: /kaggle/working/yolov8_data/train/labels.cache
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))
val: Fast image access ✓(ping: 0.0±0.0 ms, read: 693.6±747.0 MB/s, size:
111.7 KB)
val: Scanning /kaggle/working/yolov8_data/val/labels... 200 images, 0

```


backgrounds, 0 corrupt: 100% ————— 200/200 658.3it/s 0.3s0.1s
 val: New cache created: /kaggle/working/yolov8_data/val/labels.cache
 Plotting labels to /kaggle/working/runs/stenosis_segmentation/labels.jpg...
 optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937'
 and determining best 'optimizer', 'lr0' and 'momentum' automatically...
 optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 66
 weight(decay=0.0), 77 weight(decay=0.0005), 76 bias(decay=0.0)
 Image sizes 640 train, 640 val
 Using 4 dataloader workers
 Logging results to /kaggle/working/runs/stenosis_segmentation
 Starting training for 50 epochs...

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.5it/s	2.0s0.3s	
		all	200	406	0.00293	0.433
0.000733	0.00292	0.431	0.00223	0.000647		0.00211

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s0.3s	
		all	200	406	0.00379	0.367
0.00399	0.00225	0.286	0.00811	0.00205		0.0139

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.6it/s	2.0s0.3s	
		all	200	406	0.114	0.0911
0.0105	0.0619	0.0419	0.0161	0.00386		0.0358

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.6it/s	2.0s0.3s	
		all	200	406	0.0882	0.121
0.0158	0.122	0.108	0.0526	0.0136		0.0483

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.9s0.3s	
		all	200	406	0.291	0.0911
0.0257	0.3	0.0911	0.0874	0.0206		0.0854

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size						

ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.6it/s	2.0s	0.3s
		all	200	406	0.0953	0.158
0.0152	0.142	0.14	0.0745	0.0194		0.0526

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.197	0.163	
0.0305	0.225	0.17	0.115	0.0309		0.0956	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.5it/s	2.0s	0.3s	
		all	200	406	0.175	0.195	
0.043	0.195	0.217	0.13	0.0392		0.12	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.6it/s	1.9s	0.3s	
		all	200	406	0.189	0.175	
0.0373	0.209	0.172	0.131	0.0382		0.115	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.275	0.145	
0.0412	0.308	0.162	0.145	0.0421		0.121	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s	0.3s	
		all	200	406	0.233	0.232	
0.0498	0.256	0.234	0.157	0.0465		0.14	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.203	0.2	
0.0511	0.231	0.209	0.143	0.043		0.133	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
------	-------	---------	----------	----------	----------	----------	-----------

ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.6it/s	2.0s	0.3s
		all	200	406	0.238	0.236
0.0631	0.283	0.251	0.205	0.0651		0.167

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.213	0.261	
0.0721	0.277	0.268	0.215	0.0706		0.182	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.205	0.276	
0.0599	0.241	0.291	0.17	0.0548		0.15	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.9s	0.3s	
		all	200	406	0.2	0.278	
0.0553	0.242	0.283	0.184	0.056		0.158	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s	0.3s	
		all	200	406	0.242	0.297	
0.0784	0.27	0.319	0.226	0.0789		0.204	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.288	0.273	
0.0832	0.321	0.305	0.251	0.0813		0.219	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s	0.3s	
		all	200	406	0.26	0.278	
0.0771	0.267	0.308	0.214	0.071		0.194	

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							

ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s0.3s	
		all	200	406	0.294	0.313
0.0829	0.336	0.32	0.262	0.086		0.223

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s0.3s		
		all	200	406	0.273	0.256	0.207
0.0758	0.349	0.3	0.27	0.0875			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.7it/s	1.9s0.3s		
		all	200	406	0.278	0.291	0.227
0.08	0.355	0.338	0.28	0.0857			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s0.3s		
		all	200	406	0.268	0.293	0.193
0.0723	0.292	0.308	0.209	0.0697			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s0.3s		
		all	200	406	0.316	0.286	0.253
0.0971	0.384	0.323	0.302	0.097			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s0.3s		
		all	200	406	0.342	0.298	0.244
0.0937	0.403	0.323	0.289	0.0935			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.7s0.3s		
		all	200	406	0.292	0.335	0.241
0.0905	0.374	0.335	0.29	0.0955			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
------	-------	---------	----------	----------	----------	----------	-----------

ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	4.1it/s	1.7s	0.3s
		all	200	406	0.311	0.333
0.102	0.386	0.345	0.315	0.109		0.251

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s	0.3s	
		all	200	406	0.293	0.291	
0.0815	0.362	0.328	0.275	0.088		0.222	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s	0.3s	
		all	200	406	0.267	0.379	
0.105	0.36	0.367	0.338	0.112		0.269	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s	0.3s	
		all	200	406	0.312	0.36	
0.105	0.373	0.419	0.334	0.108		0.279	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s	0.3s	
		all	200	406	0.267	0.36	
0.0925	0.378	0.399	0.324	0.104		0.249	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.8it/s	1.8s	0.3s	
		all	200	406	0.307	0.337	
0.108	0.364	0.37	0.334	0.107		0.272	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s	0.3s	
		all	200	406	0.314	0.373	
0.111	0.378	0.392	0.338	0.116		0.289	

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
------	-------	---------	----------	----------	----------	----------	-----------

ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s0.3s	
		all	200	406	0.287	0.35
0.104	0.346	0.399	0.306	0.101		0.258

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s0.3s		
		all	200	406	0.303	0.352	0.264
0.1	0.391	0.361	0.336	0.108			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s0.3s		
		all	200	406	0.313	0.355	0.27
0.102	0.371	0.401	0.336	0.112			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s0.3s		
		all	200	406	0.311	0.401	0.269
0.104	0.364	0.409	0.32	0.104			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	3.9it/s	1.8s0.3s		
		all	200	406	0.324	0.397	0.268
0.0989	0.394	0.443	0.334	0.105			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s0.3s		
		all	200	406	0.318	0.401	0.277
0.107	0.382	0.446	0.35	0.115			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s0.3s		
		all	200	406	0.311	0.429	0.282
0.104	0.383	0.449	0.349	0.113			

Closing dataloader mosaic

albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',

num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	4.0it/s 1.8s	0.3s
		all	200	406	0.334	0.377	0.287
0.114	0.411	0.408	0.357	0.121			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	3.9it/s 1.8s	0.3s
		all	200	406	0.347	0.426	0.305
0.115	0.437	0.397	0.363	0.123			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	3.9it/s 1.8s	0.3s
		all	200	406	0.341	0.397	0.308
0.119	0.424	0.414	0.376	0.125			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	4.0it/s 1.8s	0.3s
		all	200	406	0.354	0.435	0.325
0.125	0.387	0.458	0.368	0.13			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	4.1it/s 1.7s	0.3s
		all	200	406	0.333	0.431	0.307
0.117	0.39	0.429	0.365	0.125			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	3.9it/s 1.8s	0.3s
		all	200	406	0.363	0.414	0.326
0.128	0.418	0.453	0.384	0.134			

	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
Size							
ages	Instances		Box(P	R	mAP50	mAP50-95)	Mask(P
R	mAP50	mAP50-95): 100%	—————		7/7	4.1it/s 1.7s	0.3s

		all	200	406	0.411	0.399	0.338
0.128	0.457	0.419	0.39	0.136			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.7s	0.3s	
	all	200	406	0.381	0.413	0.331	
0.126	0.463	0.456	0.4	0.137			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.1it/s	1.7s	0.3s	
	all	200	406	0.392	0.409	0.337	
0.132	0.48	0.414	0.409	0.14			

Size	Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df1_loss	Instances
ages	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	
R	mAP50	mAP50-95): 100%	7/7	4.0it/s	1.8s	0.3s	
	all	200	406	0.386	0.389	0.34	
0.131	0.455	0.411	0.402	0.139			

50 epochs completed in 0.236 hours.

Optimizer stripped from

/kaggle/working/runs/stenosis_segmentation/weights/last.pt, 6.8MB

Optimizer stripped from

/kaggle/working/runs/stenosis_segmentation/weights/best.pt, 6.8MB

Validating /kaggle/working/runs/stenosis_segmentation/weights/best.pt...

Ultralytics 8.3.193 Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

YOLOv8n-seg summary (fused): 85 layers, 3,258,259 parameters, 0 gradients, 11.3 GFLOPs

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

WARNING ⚠ Limiting validation plots to first 50 items per image for speed...

[illegible]

```

WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
ages  Instances      Box(P          R    mAP50  mAP50-95)    Mask(P
R      mAP50  mAP50-95): 100% -----7/7 2.0it/s 3.5s0.4s

```

/usr/local/lib/python3.11/dist-packages/matplotlib/colors.py:721:

RuntimeWarning: invalid value encountered in less

```
xa[xa < 0] = -1
```

/usr/local/lib/python3.11/dist-packages/matplotlib/colors.py:721:

RuntimeWarning: invalid value encountered in less

```
xa[xa < 0] = -1
```

```

                                all      200      406      0.391      0.409      0.336
0.132      0.474      0.414      0.409      0.141

```

Speed: 0.2ms preprocess, 2.9ms inference, 0.0ms loss, 3.8ms postprocess per image

Results saved to /kaggle/working/runs/stenosis_segmentation

Ultralytics 8.3.193 [?] Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

YOLOv8n-seg summary (fused): 85 layers, 3,258,259 parameters, 0 gradients, 11.3 GFLOPs

val: Fast image access ✓(ping: 0.0±0.0 ms, read: 1691.3±1089.7 MB/s, size: 121.1 KB)

val: Scanning /kaggle/working/yolov8_data/val/labels.cache... 200 images, 0 backgrounds, 0 corrupt: 100% -----200/200 352.8Kit/s 0.0s

```

WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...

```

```

ages  Instances      Box(P          R    mAP50  mAP50-95)    Mask(P
R      mAP50  mAP50-95): 8% - -----1/13 0.4it/s 0.8s<33.7sWARNING [?]

```

Limiting validation plots to first 50 items per image for speed...

```

WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...

```

```

WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
ages  Instances      Box(P          R      mAP50  mAP50-95)      Mask(P
R      mAP50  mAP50-95): 15%  —————  2/13 0.7it/s 1.4s<14.8sWARNING [?]
Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
WARNING [?] Limiting validation plots to first 50 items per image for speed...
ages  Instances      Box(P          R      mAP50  mAP50-95)      Mask(P
R      mAP50  mAP50-95): 100%  ————— 13/13 3.2it/s 4.0s0.2s

```

/usr/local/lib/python3.11/dist-packages/matplotlib/colors.py:721:

RuntimeWarning: invalid value encountered in less

```
xa[xa < 0] = -1
```

/usr/local/lib/python3.11/dist-packages/matplotlib/colors.py:721:

RuntimeWarning: invalid value encountered in less

```
xa[xa < 0] = -1
```

```

                                all          200          406          0.387          0.409          0.337
0.131          0.481          0.411          0.405          0.139

```

Speed: 1.7ms preprocess, 5.7ms inference, 0.0ms loss, 2.6ms postprocess per image

Results saved to /kaggle/working/runs/stenosis_segmentation2

image 1/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/1.png:

640x640 1 stenosis, 9.4ms

image 2/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/10.png:

640x640 (no detections), 9.5ms

image 3/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/100.png:

640x640 1 stenosis, 9.4ms

image 4/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/101.png:

640x640 3 stenosis, 9.4ms

image 5/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/102.png:

640x640 1 stenosis, 9.5ms

image 6/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/103.png:

640x640 1 stenosis, 9.4ms

image 7/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/104.png:

640x640 2 stenosis, 9.4ms

image 8/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/105.png:

640x640 4 stenosis, 9.5ms

image 9/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/106.png: 640x640 2 stenosis, 9.4ms
image 10/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/107.png: 640x640 4 stenosis, 9.4ms
image 11/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/108.png: 640x640 3 stenosis, 9.4ms
image 12/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/109.png: 640x640 2 stenosis, 9.4ms
image 13/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/11.png: 640x640 4 stenosis, 9.4ms
image 14/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/110.png: 640x640 1 stenosis, 9.4ms
image 15/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/111.png: 640x640 1 stenosis, 9.4ms
image 16/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/112.png: 640x640 2 stenosis, 9.4ms
image 17/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/113.png: 640x640 2 stenosis, 9.4ms
image 18/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/114.png: 640x640 2 stenosis, 9.5ms
image 19/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/115.png: 640x640 (no detections), 7.9ms
image 20/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/116.png: 640x640 2 stenosis, 7.9ms
image 21/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/117.png: 640x640 1 stenosis, 7.9ms
image 22/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/118.png: 640x640 (no detections), 7.9ms
image 23/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/119.png: 640x640 (no detections), 7.9ms
image 24/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/12.png: 640x640 3 stenosis, 8.2ms
image 25/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/120.png: 640x640 1 stenosis, 8.0ms
image 26/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/121.png: 640x640 1 stenosis, 8.1ms
image 27/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/122.png: 640x640 5 stenosis, 7.9ms
image 28/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/123.png: 640x640 1 stenosis, 7.4ms
image 29/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/124.png: 640x640 5 stenosis, 7.5ms
image 30/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/125.png: 640x640 1 stenosis, 7.4ms
image 31/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/126.png: 640x640 3 stenosis, 7.5ms
image 32/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/127.png: 640x640 1 stenosis, 7.4ms
image 33/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/128.png: 640x640 2 stenosis, 7.5ms

image 34/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/129.png: 640x640 2 stenosis, 7.4ms
image 35/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/13.png: 640x640 1 stenosis, 7.5ms
image 36/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/130.png: 640x640 8 stenosis, 7.5ms
image 37/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/131.png: 640x640 (no detections), 7.3ms
image 38/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/132.png: 640x640 5 stenosis, 7.3ms
image 39/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/133.png: 640x640 2 stenosis, 7.3ms
image 40/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/134.png: 640x640 (no detections), 7.2ms
image 41/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/135.png: 640x640 (no detections), 7.2ms
image 42/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/136.png: 640x640 2 stenosis, 7.3ms
image 43/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/137.png: 640x640 1 stenosis, 7.3ms
image 44/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/138.png: 640x640 (no detections), 7.4ms
image 45/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/139.png: 640x640 4 stenosis, 7.3ms
image 46/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/14.png: 640x640 3 stenosis, 7.3ms
image 47/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/140.png: 640x640 (no detections), 7.4ms
image 48/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/141.png: 640x640 1 stenosis, 7.1ms
image 49/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/142.png: 640x640 (no detections), 7.7ms
image 50/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/143.png: 640x640 2 stenosis, 7.0ms
image 51/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/144.png: 640x640 1 stenosis, 7.0ms
image 52/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/145.png: 640x640 2 stenosis, 7.2ms
image 53/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/146.png: 640x640 2 stenosis, 7.1ms
image 54/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/147.png: 640x640 3 stenosis, 7.7ms
image 55/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/148.png: 640x640 (no detections), 7.1ms
image 56/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/149.png: 640x640 1 stenosis, 7.2ms
image 57/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/15.png: 640x640 2 stenosis, 7.2ms
image 58/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/150.png: 640x640 1 stenosis, 7.1ms

image 59/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/151.png: 640x640 2 stenosis, 7.2ms
image 60/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/152.png: 640x640 1 stenosis, 7.1ms
image 61/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/153.png: 640x640 2 stenosis, 7.1ms
image 62/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/154.png: 640x640 2 stenosis, 7.1ms
image 63/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/155.png: 640x640 1 stenosis, 8.0ms
image 64/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/156.png: 640x640 2 stenosis, 7.9ms
image 65/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/157.png: 640x640 (no detections), 7.4ms
image 66/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/158.png: 640x640 1 stenosis, 7.5ms
image 67/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/159.png: 640x640 2 stenosis, 7.1ms
image 68/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/16.png: 640x640 2 stenosis, 7.2ms
image 69/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/160.png: 640x640 2 stenosis, 7.6ms
image 70/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/161.png: 640x640 1 stenosis, 7.3ms
image 71/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/162.png: 640x640 4 stenosis, 7.8ms
image 72/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/163.png: 640x640 (no detections), 7.7ms
image 73/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/164.png: 640x640 3 stenosis, 7.3ms
image 74/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/165.png: 640x640 (no detections), 7.4ms
image 75/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/166.png: 640x640 2 stenosis, 7.3ms
image 76/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/167.png: 640x640 1 stenosis, 7.4ms
image 77/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/168.png: 640x640 3 stenosis, 7.3ms
image 78/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/169.png: 640x640 1 stenosis, 7.1ms
image 79/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/17.png: 640x640 1 stenosis, 7.4ms
image 80/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/170.png: 640x640 4 stenosis, 7.2ms
image 81/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/171.png: 640x640 (no detections), 7.7ms
image 82/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/172.png: 640x640 1 stenosis, 7.1ms
image 83/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/173.png: 640x640 1 stenosis, 7.1ms

image 84/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/174.png: 640x640 1 stenosis, 7.0ms
image 85/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/175.png: 640x640 2 stenosis, 7.1ms
image 86/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/176.png: 640x640 2 stenosis, 7.0ms
image 87/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/177.png: 640x640 5 stenosis, 7.1ms
image 88/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/178.png: 640x640 1 stenosis, 7.4ms
image 89/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/179.png: 640x640 (no detections), 7.7ms
image 90/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/18.png: 640x640 2 stenosis, 7.0ms
image 91/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/180.png: 640x640 1 stenosis, 7.0ms
image 92/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/181.png: 640x640 (no detections), 7.2ms
image 93/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/182.png: 640x640 2 stenosis, 7.2ms
image 94/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/183.png: 640x640 3 stenosis, 7.5ms
image 95/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/184.png: 640x640 1 stenosis, 7.9ms
image 96/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/185.png: 640x640 (no detections), 7.2ms
image 97/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/186.png: 640x640 2 stenosis, 7.0ms
image 98/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/187.png: 640x640 2 stenosis, 7.1ms
image 99/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/188.png: 640x640 2 stenosis, 7.4ms
image 100/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/189.png: 640x640 3 stenosis, 7.0ms
image 101/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/19.png: 640x640 1 stenosis, 7.3ms
image 102/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/190.png: 640x640 2 stenosis, 7.2ms
image 103/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/191.png: 640x640 5 stenosis, 7.2ms
image 104/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/192.png: 640x640 1 stenosis, 7.1ms
image 105/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/193.png: 640x640 1 stenosis, 7.4ms
image 106/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/194.png: 640x640 2 stenosis, 7.5ms
image 107/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/195.png: 640x640 3 stenosis, 8.2ms
image 108/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/196.png: 640x640 2 stenosis, 7.9ms

image 109/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/197.png: 640x640 1 stenosis, 7.6ms
image 110/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/198.png: 640x640 3 stenosis, 7.6ms
image 111/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/199.png: 640x640 2 stenosis, 8.1ms
image 112/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/2.png: 640x640 2 stenosis, 7.8ms
image 113/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/20.png: 640x640 2 stenosis, 7.7ms
image 114/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/200.png: 640x640 1 stenosis, 8.0ms
image 115/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/201.png: 640x640 1 stenosis, 8.1ms
image 116/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/202.png: 640x640 2 stenosis, 8.0ms
image 117/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/203.png: 640x640 2 stenosis, 8.0ms
image 118/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/204.png: 640x640 (no detections), 7.7ms
image 119/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/205.png: 640x640 (no detections), 8.0ms
image 120/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/206.png: 640x640 1 stenosis, 7.9ms
image 121/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/207.png: 640x640 1 stenosis, 8.0ms
image 122/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/208.png: 640x640 (no detections), 8.1ms
image 123/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/209.png: 640x640 3 stenosis, 8.0ms
image 124/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/21.png: 640x640 (no detections), 8.1ms
image 125/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/210.png: 640x640 1 stenosis, 8.1ms
image 126/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/211.png: 640x640 1 stenosis, 8.6ms
image 127/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/212.png: 640x640 2 stenosis, 8.4ms
image 128/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/213.png: 640x640 (no detections), 8.1ms
image 129/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/214.png: 640x640 2 stenosis, 8.0ms
image 130/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/215.png: 640x640 1 stenosis, 8.1ms
image 131/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/216.png: 640x640 2 stenosis, 7.9ms
image 132/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/217.png: 640x640 3 stenosis, 7.6ms
image 133/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/218.png: 640x640 1 stenosis, 8.3ms

image 134/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/219.png: 640x640 3 stenosis, 8.2ms
image 135/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/22.png: 640x640 1 stenosis, 7.7ms
image 136/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/220.png: 640x640 1 stenosis, 7.8ms
image 137/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/221.png: 640x640 2 stenosis, 7.7ms
image 138/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/222.png: 640x640 1 stenosis, 7.9ms
image 139/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/223.png: 640x640 2 stenosis, 8.0ms
image 140/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/224.png: 640x640 4 stenosis, 7.6ms
image 141/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/225.png: 640x640 2 stenosis, 8.2ms
image 142/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/226.png: 640x640 1 stenosis, 7.7ms
image 143/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/227.png: 640x640 (no detections), 8.1ms
image 144/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/228.png: 640x640 2 stenosis, 8.1ms
image 145/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/229.png: 640x640 3 stenosis, 7.8ms
image 146/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/23.png: 640x640 2 stenosis, 8.0ms
image 147/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/230.png: 640x640 3 stenosis, 8.5ms
image 148/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/231.png: 640x640 1 stenosis, 7.9ms
image 149/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/232.png: 640x640 1 stenosis, 7.9ms
image 150/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/233.png: 640x640 (no detections), 7.7ms
image 151/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/234.png: 640x640 3 stenosis, 8.2ms
image 152/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/235.png: 640x640 2 stenosis, 8.5ms
image 153/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/236.png: 640x640 (no detections), 7.9ms
image 154/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/237.png: 640x640 1 stenosis, 7.7ms
image 155/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/238.png: 640x640 1 stenosis, 8.0ms
image 156/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/239.png: 640x640 3 stenosis, 7.8ms
image 157/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/24.png: 640x640 2 stenosis, 8.0ms
image 158/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/240.png: 640x640 3 stenosis, 8.1ms

image 159/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/241.png: 640x640 3 stenosis, 7.8ms
image 160/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/242.png: 640x640 4 stenosis, 8.0ms
image 161/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/243.png: 640x640 1 stenosis, 8.3ms
image 162/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/244.png: 640x640 1 stenosis, 8.2ms
image 163/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/245.png: 640x640 (no detections), 7.9ms
image 164/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/246.png: 640x640 1 stenosis, 7.8ms
image 165/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/247.png: 640x640 3 stenosis, 8.0ms
image 166/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/248.png: 640x640 1 stenosis, 8.0ms
image 167/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/249.png: 640x640 4 stenosis, 8.7ms
image 168/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/25.png: 640x640 1 stenosis, 8.1ms
image 169/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/250.png: 640x640 1 stenosis, 8.1ms
image 170/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/251.png: 640x640 1 stenosis, 7.9ms
image 171/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/252.png: 640x640 (no detections), 8.6ms
image 172/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/253.png: 640x640 1 stenosis, 7.9ms
image 173/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/254.png: 640x640 1 stenosis, 7.9ms
image 174/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/255.png: 640x640 1 stenosis, 8.1ms
image 175/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/256.png: 640x640 1 stenosis, 9.0ms
image 176/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/257.png: 640x640 1 stenosis, 8.4ms
image 177/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/258.png: 640x640 2 stenosis, 7.5ms
image 178/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/259.png: 640x640 1 stenosis, 8.0ms
image 179/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/26.png: 640x640 2 stenosis, 7.7ms
image 180/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/260.png: 640x640 (no detections), 8.1ms
image 181/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/261.png: 640x640 2 stenosis, 7.7ms
image 182/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/262.png: 640x640 1 stenosis, 8.0ms
image 183/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/263.png: 640x640 (no detections), 8.1ms

image 184/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/264.png: 640x640 4 stenosis, 7.8ms
image 185/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/265.png: 640x640 1 stenosis, 8.2ms
image 186/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/266.png: 640x640 (no detections), 8.2ms
image 187/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/267.png: 640x640 (no detections), 9.9ms
image 188/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/268.png: 640x640 4 stenosis, 8.1ms
image 189/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/269.png: 640x640 1 stenosis, 7.9ms
image 190/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/27.png: 640x640 1 stenosis, 8.3ms
image 191/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/270.png: 640x640 2 stenosis, 8.1ms
image 192/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/271.png: 640x640 (no detections), 8.0ms
image 193/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/272.png: 640x640 1 stenosis, 8.0ms
image 194/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/273.png: 640x640 2 stenosis, 8.0ms
image 195/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/274.png: 640x640 (no detections), 8.4ms
image 196/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/275.png: 640x640 1 stenosis, 7.9ms
image 197/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/276.png: 640x640 1 stenosis, 9.3ms
image 198/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/277.png: 640x640 1 stenosis, 8.1ms
image 199/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/278.png: 640x640 1 stenosis, 8.3ms
image 200/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/279.png: 640x640 2 stenosis, 8.0ms
image 201/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/28.png: 640x640 3 stenosis, 8.2ms
image 202/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/280.png: 640x640 2 stenosis, 8.6ms
image 203/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/281.png: 640x640 2 stenosis, 7.8ms
image 204/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/282.png: 640x640 (no detections), 8.1ms
image 205/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/283.png: 640x640 (no detections), 7.8ms
image 206/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/284.png: 640x640 (no detections), 8.0ms
image 207/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/285.png: 640x640 1 stenosis, 8.4ms
image 208/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/286.png: 640x640 1 stenosis, 8.0ms

image 209/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/287.png: 640x640 2 stenosis, 8.1ms
image 210/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/288.png: 640x640 2 stenosis, 8.1ms
image 211/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/289.png: 640x640 (no detections), 8.1ms
image 212/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/29.png: 640x640 3 stenosis, 7.9ms
image 213/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/290.png: 640x640 1 stenosis, 8.1ms
image 214/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/291.png: 640x640 2 stenosis, 8.4ms
image 215/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/292.png: 640x640 2 stenosis, 8.1ms
image 216/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/293.png: 640x640 4 stenosis, 8.3ms
image 217/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/294.png: 640x640 1 stenosis, 8.1ms
image 218/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/295.png: 640x640 2 stenosis, 8.0ms
image 219/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/296.png: 640x640 2 stenosis, 7.9ms
image 220/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/297.png: 640x640 1 stenosis, 8.5ms
image 221/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/298.png: 640x640 4 stenosis, 8.3ms
image 222/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/299.png: 640x640 1 stenosis, 8.2ms
image 223/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/3.png: 640x640 2 stenosis, 8.5ms
image 224/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/30.png: 640x640 1 stenosis, 8.3ms
image 225/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/300.png: 640x640 2 stenosis, 8.1ms
image 226/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/31.png: 640x640 1 stenosis, 8.3ms
image 227/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/32.png: 640x640 2 stenosis, 8.3ms
image 228/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/33.png: 640x640 4 stenosis, 8.6ms
image 229/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/34.png: 640x640 1 stenosis, 8.1ms
image 230/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/35.png: 640x640 1 stenosis, 8.3ms
image 231/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/36.png: 640x640 2 stenosis, 8.3ms
image 232/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/37.png: 640x640 2 stenosis, 8.2ms
image 233/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/38.png: 640x640 3 stenosis, 7.7ms

image 234/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/39.png: 640x640 4 stenosis, 8.0ms
image 235/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/4.png: 640x640 1 stenosis, 8.3ms
image 236/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/40.png: 640x640 2 stenosis, 8.3ms
image 237/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/41.png: 640x640 2 stenosis, 8.4ms
image 238/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/42.png: 640x640 (no detections), 8.2ms
image 239/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/43.png: 640x640 2 stenosis, 7.9ms
image 240/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/44.png: 640x640 1 stenosis, 8.1ms
image 241/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/45.png: 640x640 2 stenosis, 8.2ms
image 242/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/46.png: 640x640 (no detections), 7.8ms
image 243/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/47.png: 640x640 2 stenosis, 7.9ms
image 244/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/48.png: 640x640 1 stenosis, 8.4ms
image 245/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/49.png: 640x640 1 stenosis, 8.1ms
image 246/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/5.png: 640x640 4 stenosis, 8.0ms
image 247/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/50.png: 640x640 (no detections), 7.9ms
image 248/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/51.png: 640x640 3 stenosis, 8.0ms
image 249/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/52.png: 640x640 2 stenosis, 9.0ms
image 250/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/53.png: 640x640 2 stenosis, 8.1ms
image 251/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/54.png: 640x640 (no detections), 8.3ms
image 252/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/55.png: 640x640 1 stenosis, 7.8ms
image 253/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/56.png: 640x640 1 stenosis, 7.8ms
image 254/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/57.png: 640x640 4 stenosis, 8.8ms
image 255/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/58.png: 640x640 (no detections), 8.3ms
image 256/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/59.png: 640x640 3 stenosis, 8.0ms
image 257/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/6.png: 640x640 (no detections), 7.8ms
image 258/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/60.png: 640x640 1 stenosis, 8.3ms

image 259/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/61.png: 640x640 2 stenosis, 8.2ms
image 260/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/62.png: 640x640 (no detections), 8.4ms
image 261/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/63.png: 640x640 1 stenosis, 7.8ms
image 262/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/64.png: 640x640 5 stenosis, 7.7ms
image 263/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/65.png: 640x640 3 stenosis, 8.1ms
image 264/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/66.png: 640x640 1 stenosis, 8.4ms
image 265/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/67.png: 640x640 2 stenosis, 8.7ms
image 266/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/68.png: 640x640 1 stenosis, 8.0ms
image 267/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/69.png: 640x640 (no detections), 7.8ms
image 268/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/7.png: 640x640 (no detections), 7.7ms
image 269/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/70.png: 640x640 2 stenosis, 7.6ms
image 270/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/71.png: 640x640 2 stenosis, 7.8ms
image 271/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/72.png: 640x640 1 stenosis, 7.5ms
image 272/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/73.png: 640x640 3 stenosis, 7.4ms
image 273/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/74.png: 640x640 (no detections), 7.4ms
image 274/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/75.png: 640x640 (no detections), 7.4ms
image 275/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/76.png: 640x640 2 stenosis, 7.4ms
image 276/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/77.png: 640x640 2 stenosis, 7.8ms
image 277/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/78.png: 640x640 4 stenosis, 7.5ms
image 278/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/79.png: 640x640 6 stenosis, 7.3ms
image 279/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/8.png: 640x640 1 stenosis, 7.2ms
image 280/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/80.png: 640x640 2 stenosis, 7.4ms
image 281/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/81.png: 640x640 (no detections), 7.6ms
image 282/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/82.png: 640x640 2 stenosis, 7.2ms
image 283/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/83.png: 640x640 1 stenosis, 7.5ms

```

image 284/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/84.png: 640x640 (no detections), 8.1ms
image 285/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/85.png: 640x640 1 stenosis, 7.3ms
image 286/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/86.png: 640x640 1 stenosis, 7.5ms
image 287/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/87.png: 640x640 2 stenosis, 7.3ms
image 288/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/88.png: 640x640 (no detections), 7.3ms
image 289/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/89.png: 640x640 4 stenosis, 7.6ms
image 290/300 /kaggle/input/arcade-dataset/arcade/stenosis/test/images/9.png:
640x640 2 stenosis, 7.3ms
image 291/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/90.png: 640x640 3 stenosis, 7.5ms
image 292/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/91.png: 640x640 1 stenosis, 7.1ms
image 293/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/92.png: 640x640 2 stenosis, 7.1ms
image 294/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/93.png: 640x640 1 stenosis, 7.1ms
image 295/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/94.png: 640x640 1 stenosis, 7.2ms
image 296/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/95.png: 640x640 3 stenosis, 7.1ms
image 297/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/96.png: 640x640 3 stenosis, 7.5ms
image 298/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/97.png: 640x640 1 stenosis, 7.1ms
image 299/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/98.png: 640x640 1 stenosis, 7.3ms
image 300/300 /kaggle/input/arcade-
dataset/arcade/stenosis/test/images/99.png: 640x640 1 stenosis, 7.1ms
Speed: 2.8ms preprocess, 7.9ms inference, 1.7ms postprocess per image at
shape (1, 3, 640, 640)
Results saved to /kaggle/working/runs/test_predictions
Training and inference completed. Results saved in /kaggle/working/runs.

```

```

import os
import cv2
import matplotlib.pyplot as plt

```

```

folder_path = "/kaggle/working/runs/stenosis_segmentation2"

```

```

image_files = [f for f in os.listdir(folder_path) if
f.lower().endswith(('.png', '.jpg', '.jpeg'))]

```

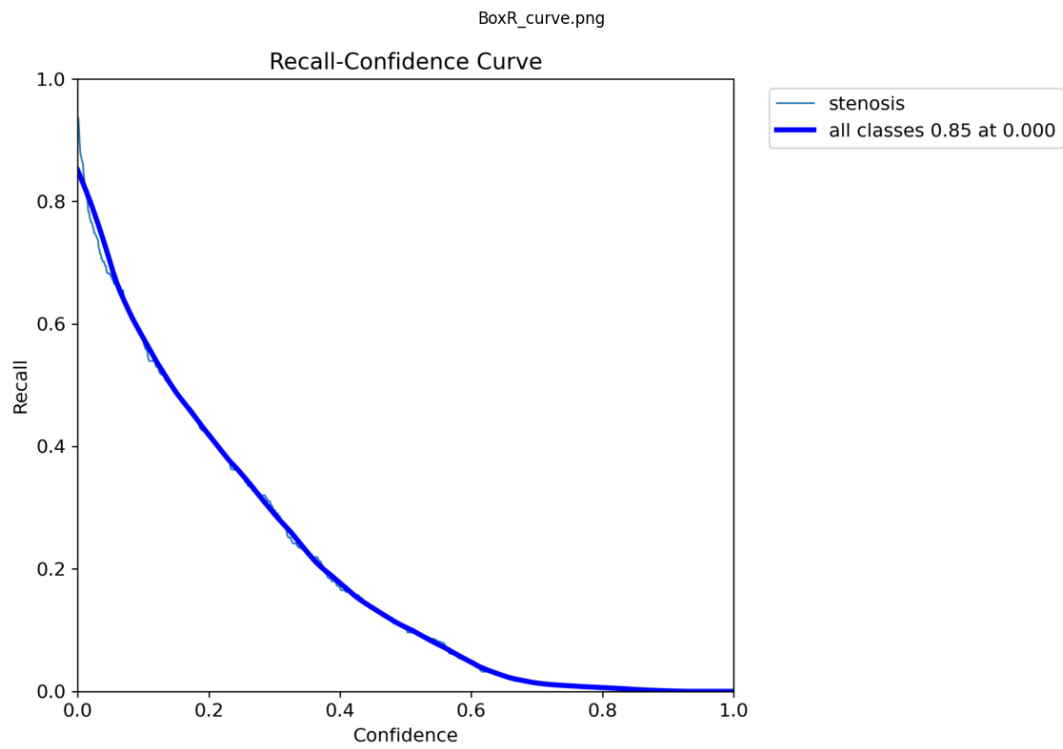
```

for img_file in image_files:
    img_path = os.path.join(folder_path, img_file)

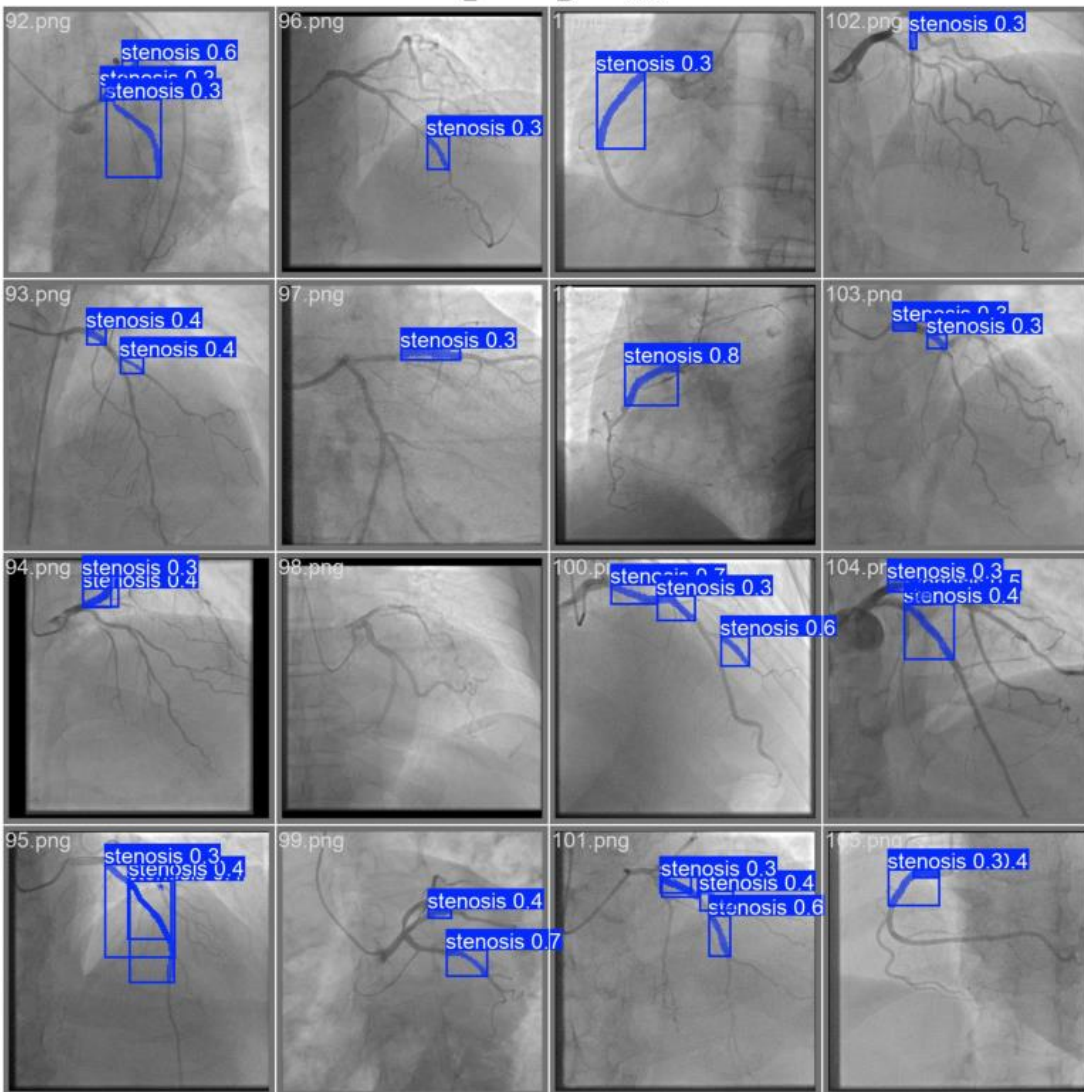
```

```
img = cv2.imread(img_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

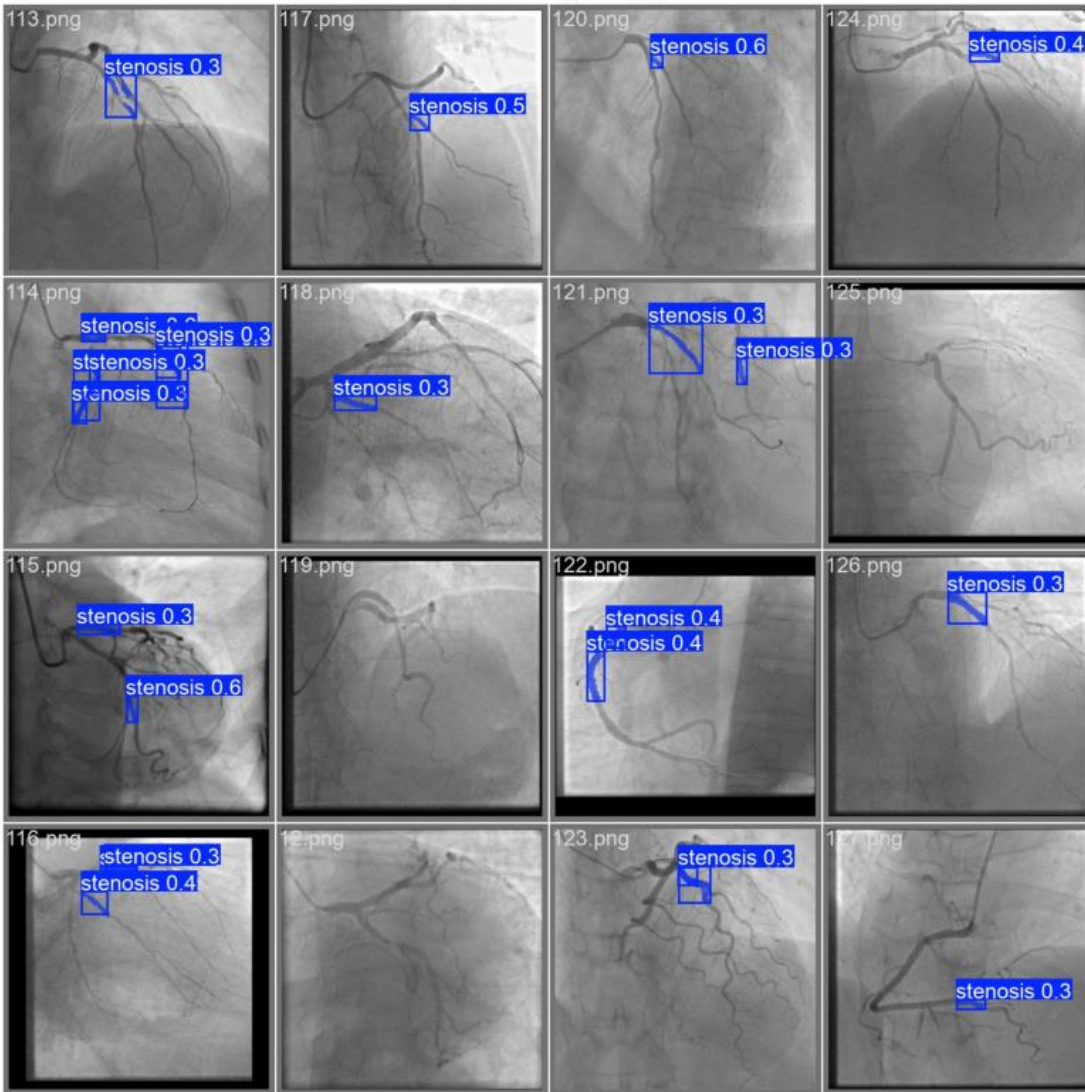
```
plt.figure(figsize=(15, 10))
plt.imshow(img)
plt.title(img_file)
plt.axis("off")
plt.show()
```



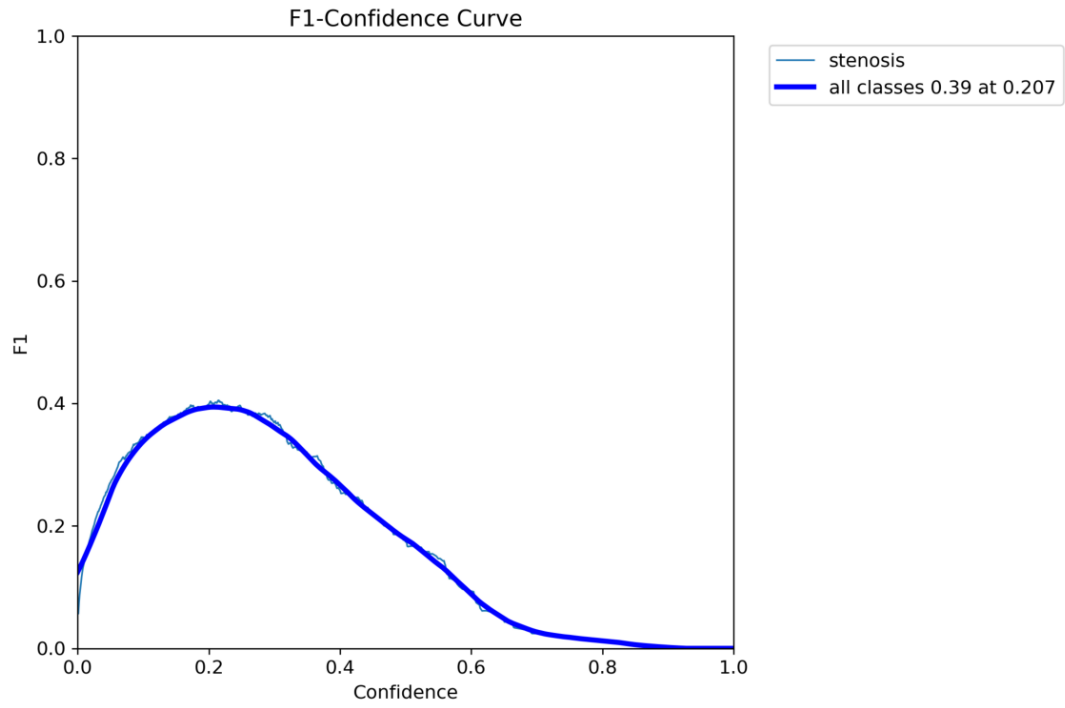
val_batch0_pred.jpg



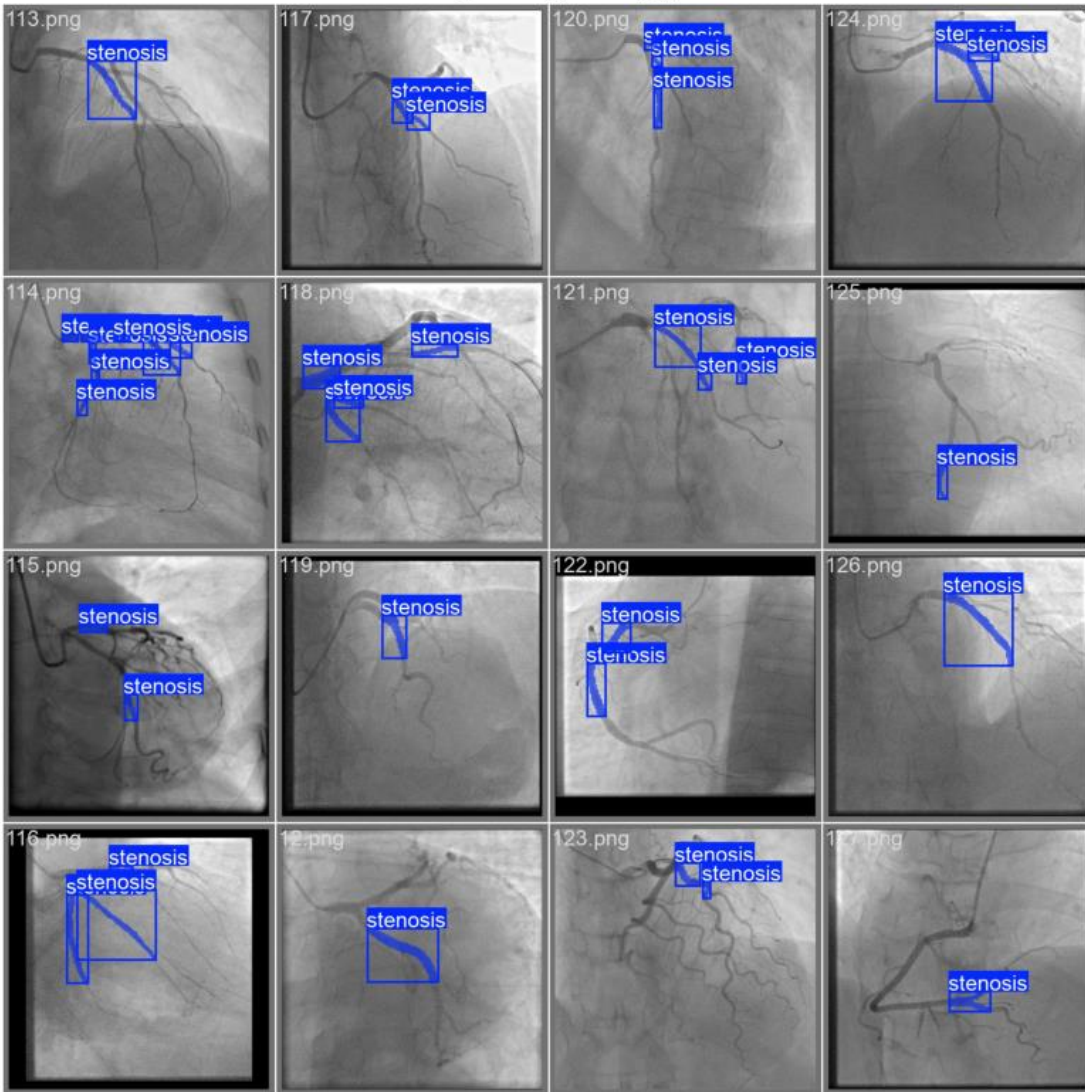
val_batch1_pred.jpg

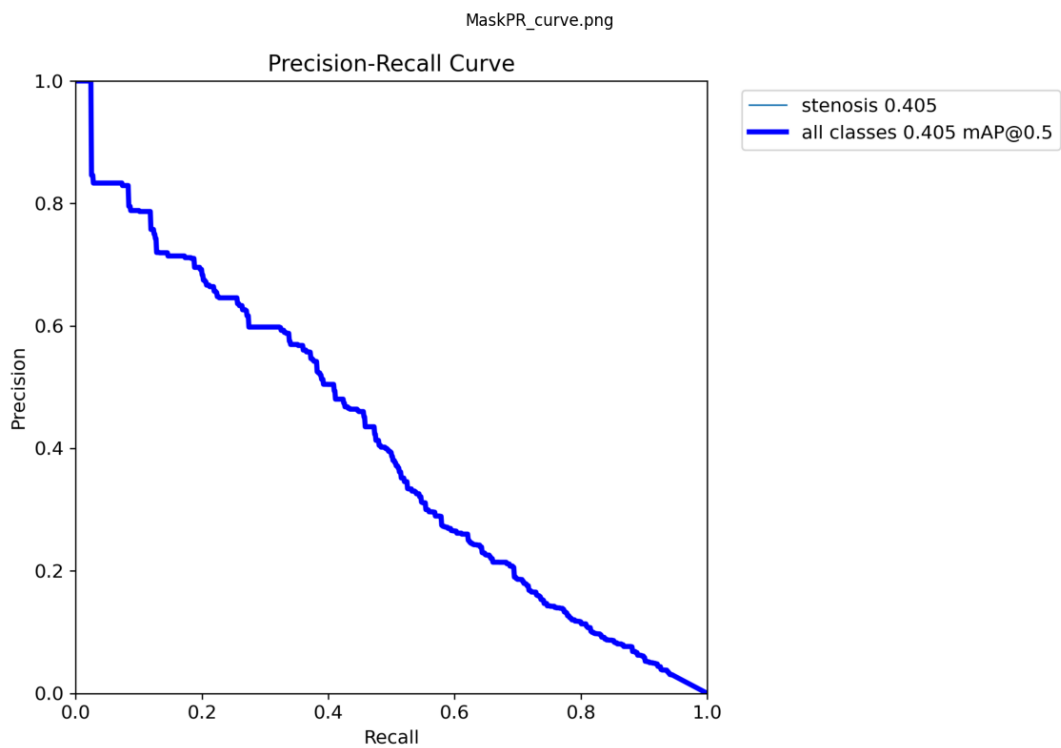
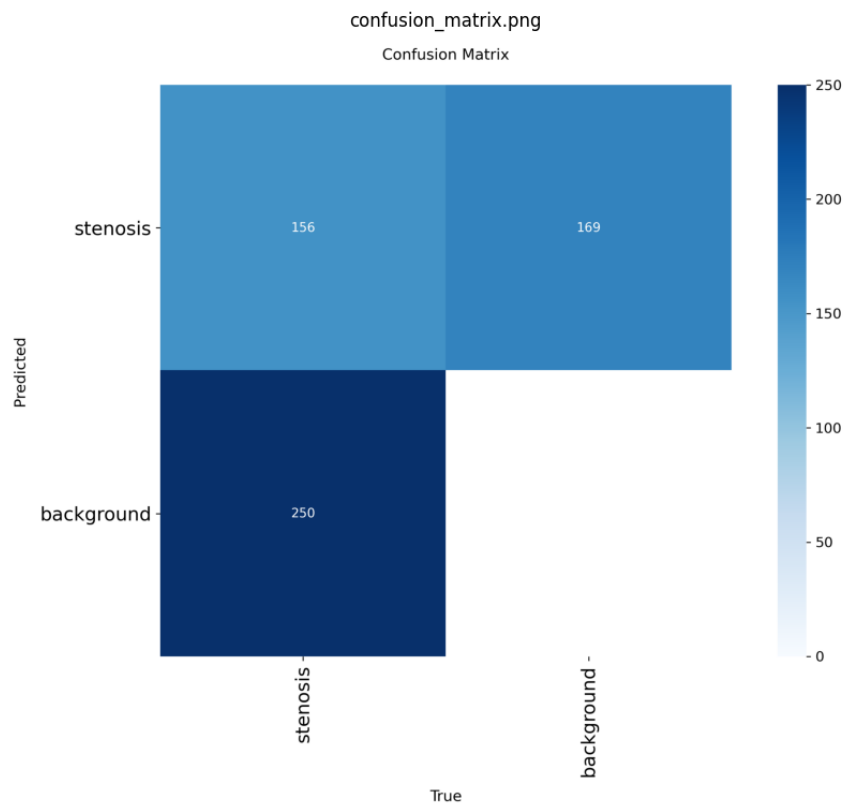


BoxF1_curve.png

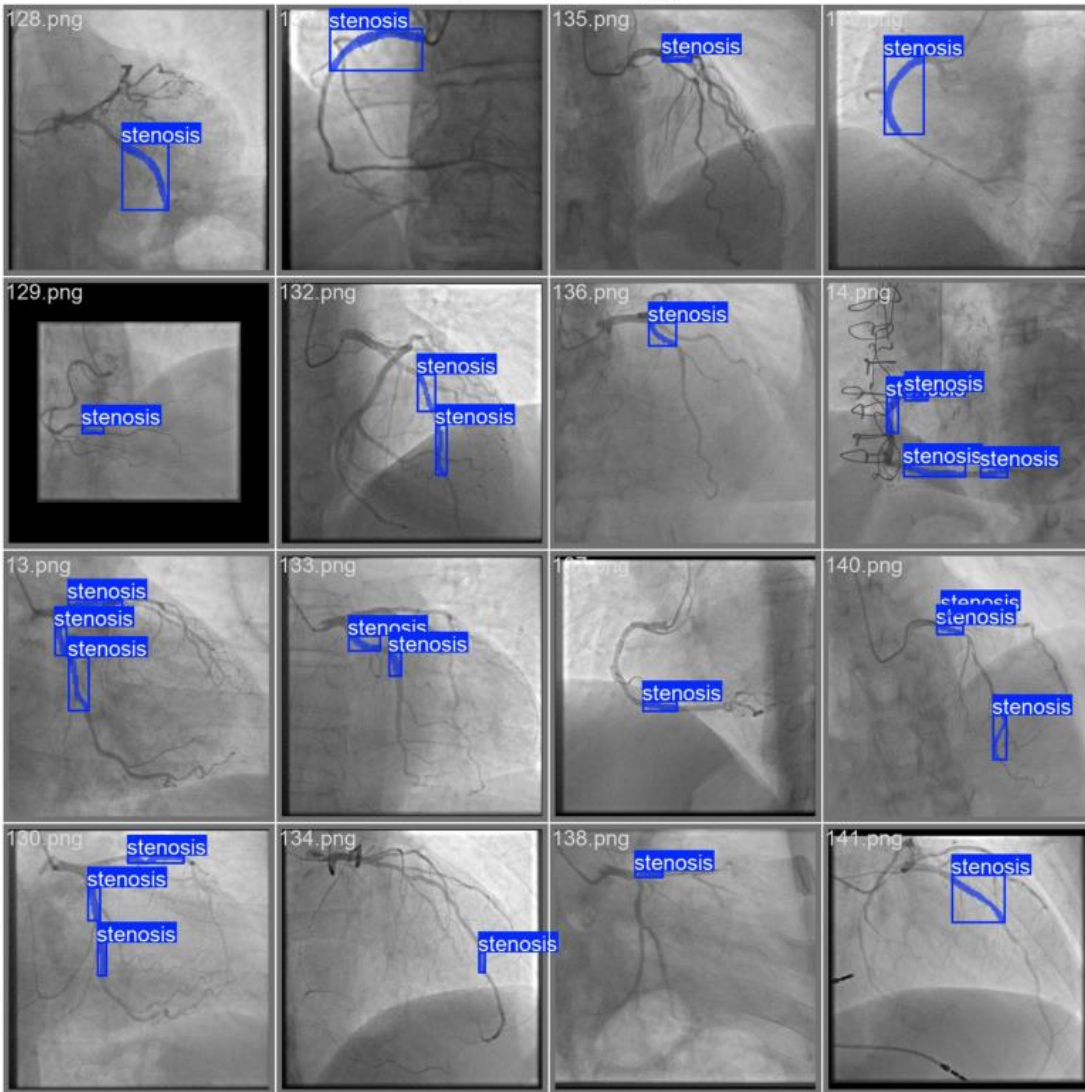


val_batch1_labels.jpg

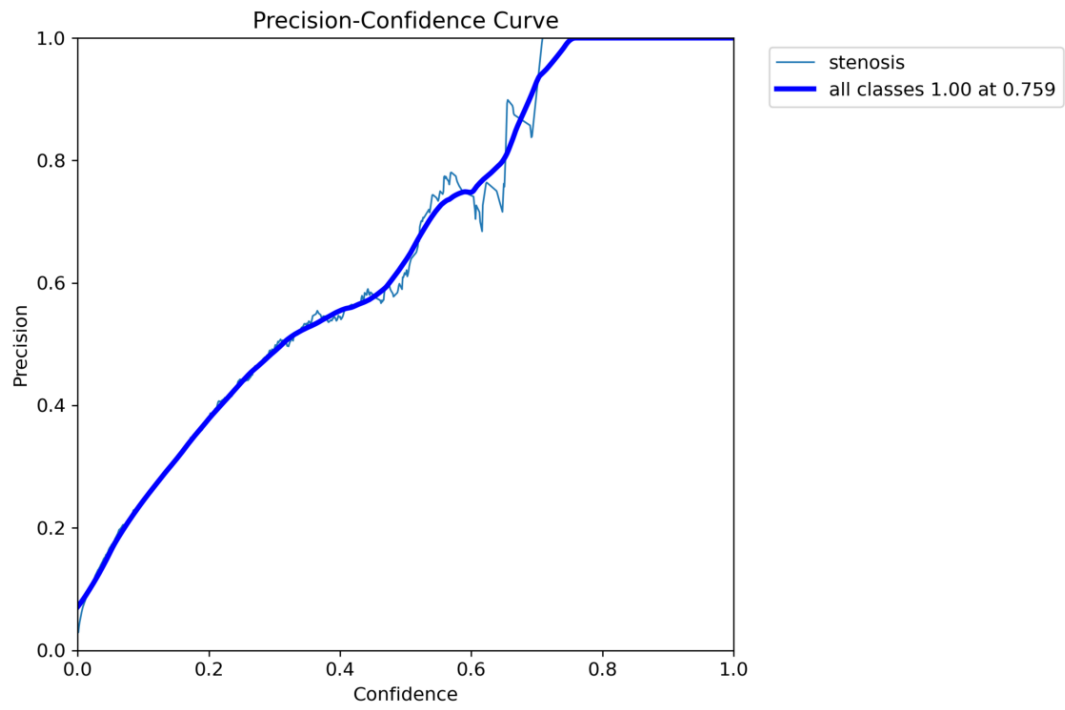




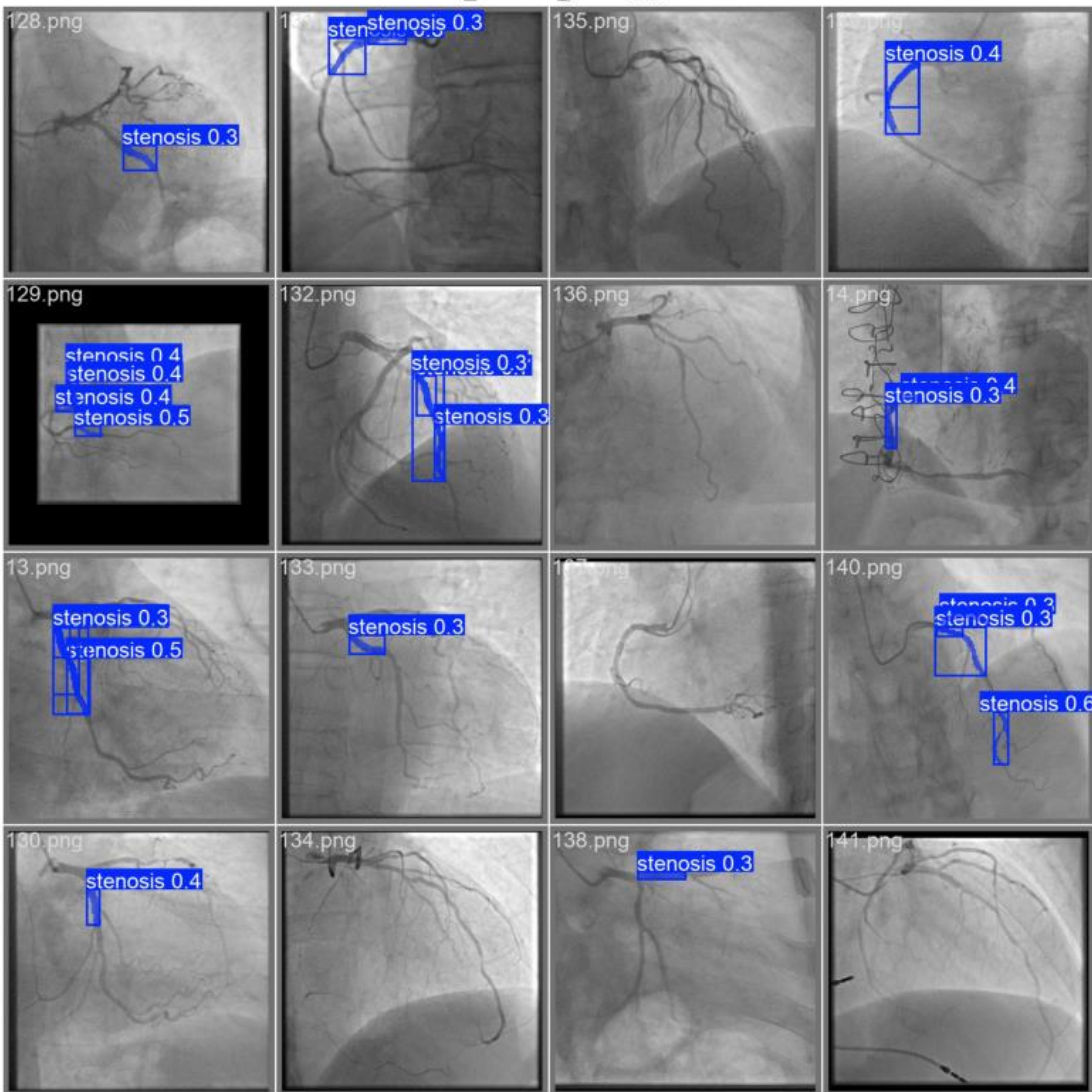
val_batch2_labels.jpg

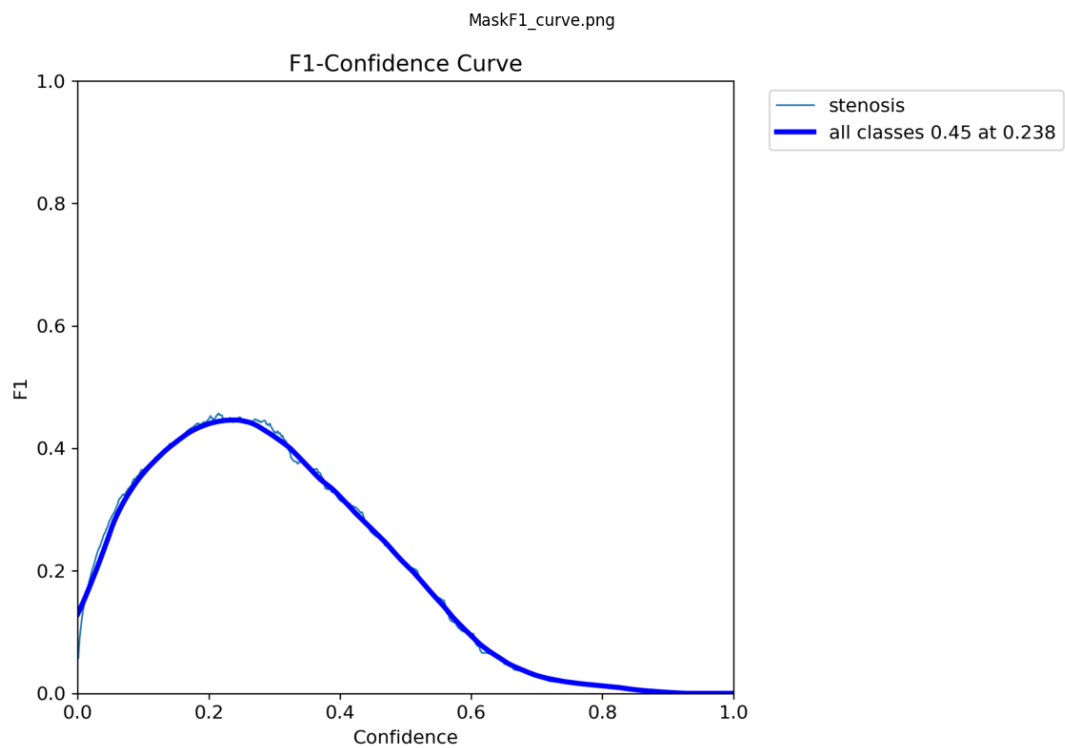
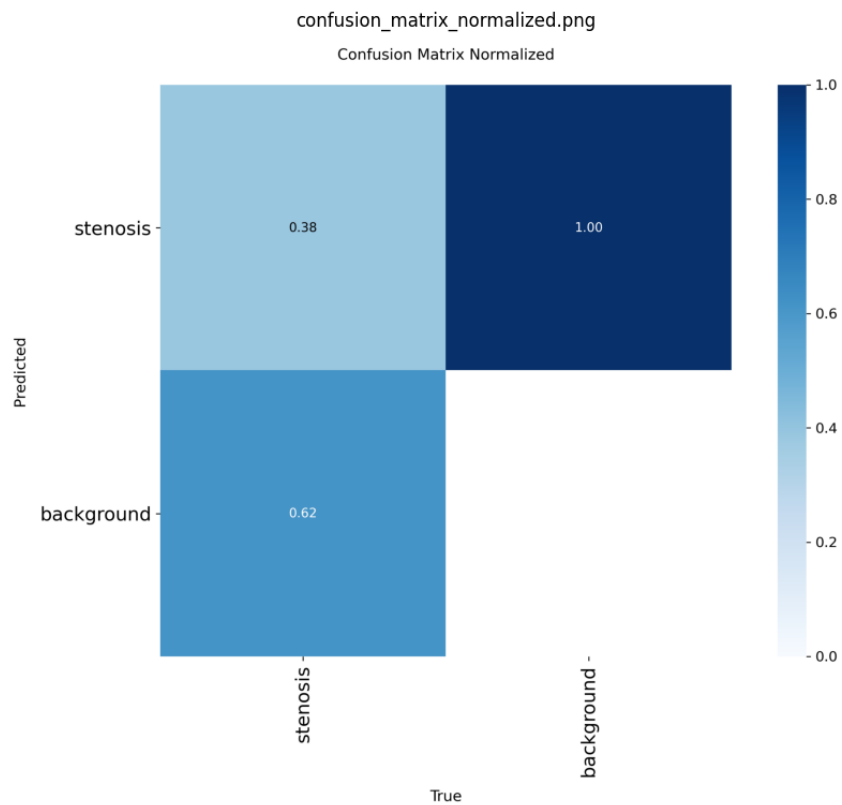


BoxP_curve.png

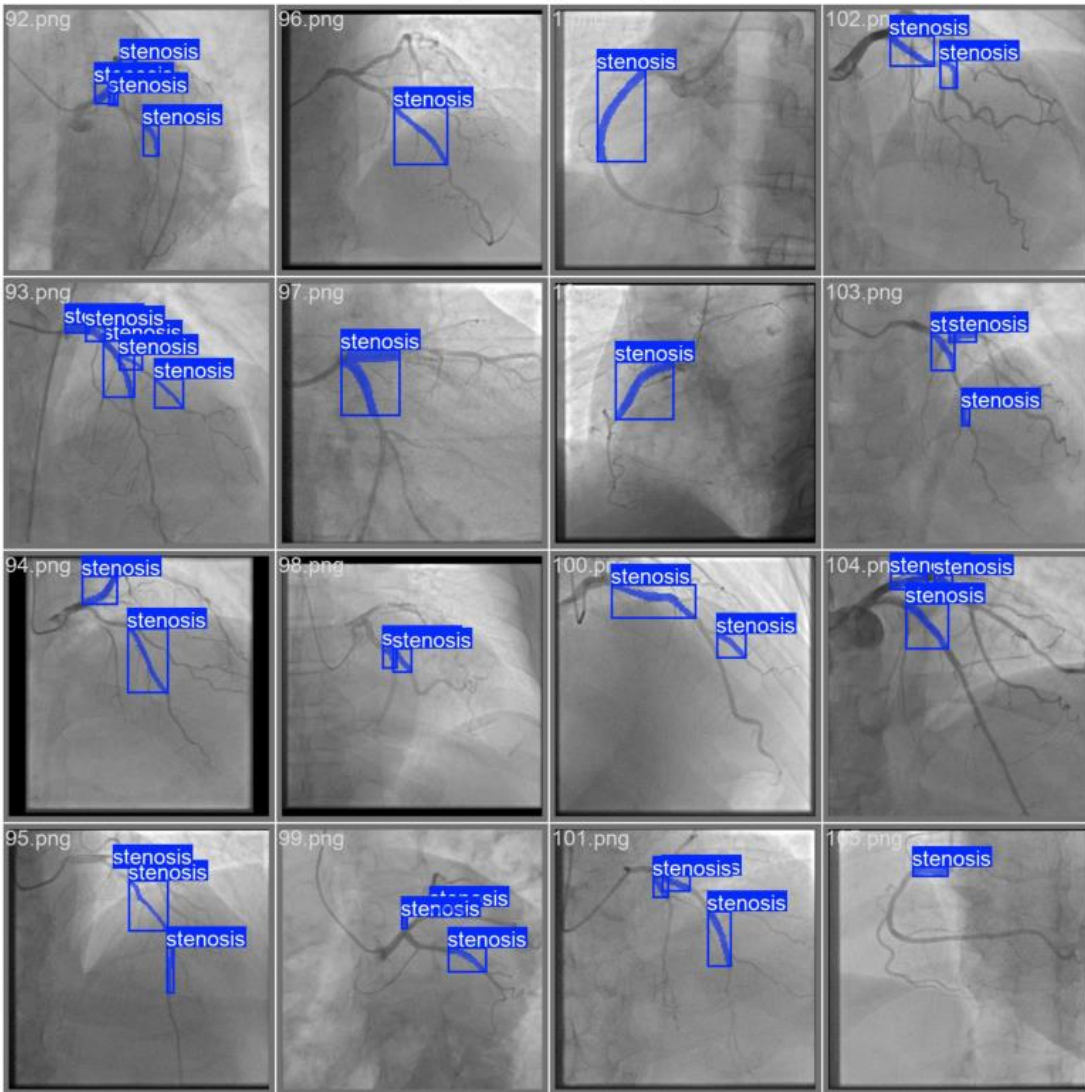


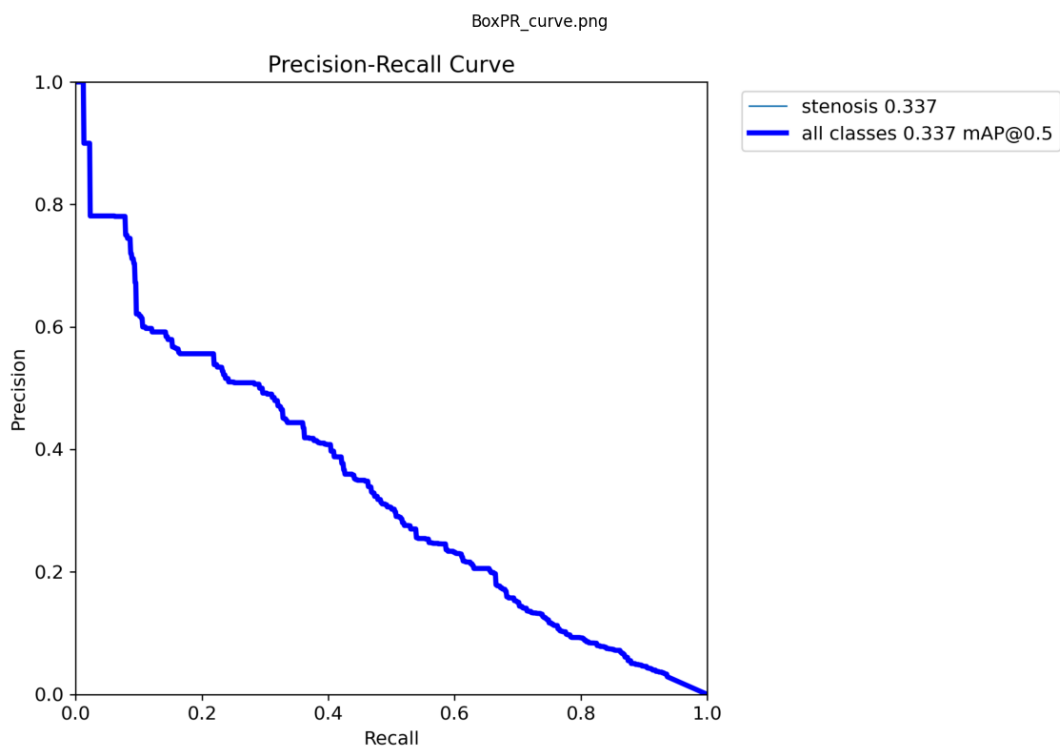
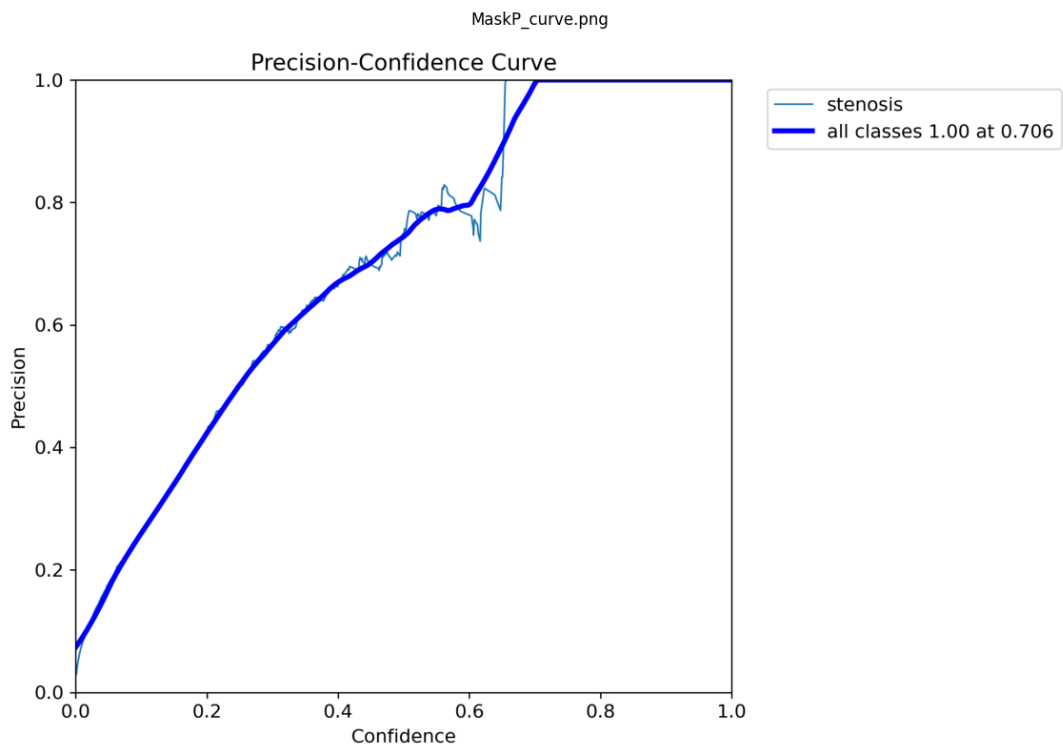
val_batch2_pred.jpg

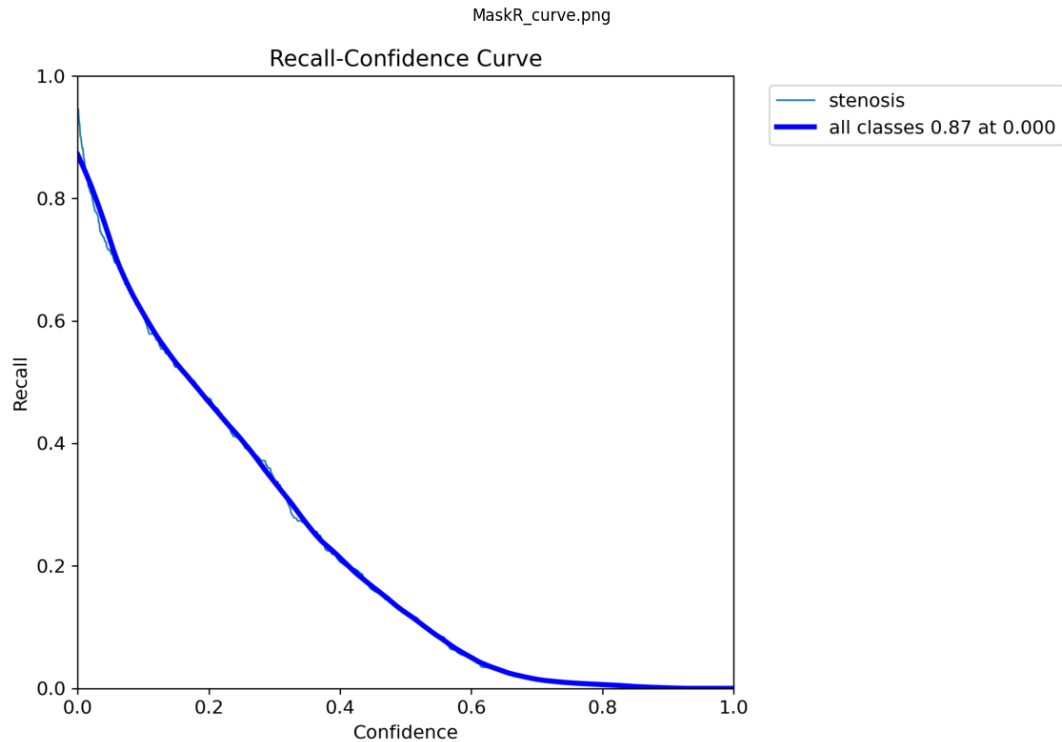




val_batch0_labels.jpg







```
import random

older_path = "/kaggle/working/runs/test_predictions"

image_files = [f for f in os.listdir(folder_path) if
f.lower().endswith(('.png', '.jpg', '.jpeg'))]

sample_files = random.sample(image_files, min(20, len(image_files)))

plt.figure(figsize=(15, 12))
for i, img_file in enumerate(sample_files, 1):
    img_path = os.path.join(folder_path, img_file)
    img = cv2.imread(img_path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    plt.subplot(5, 4, i)
    plt.imshow(img)
    plt.title(img_file, fontsize=8)
    plt.axis("off")

plt.tight_layout()
plt.show()
```

